Precipitation Extremes, Rural Livelihoods, and Fertility in China

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Abstract

This study examines how extreme precipitation affects fertility in rural China. We estimate that an extremely dry or wet year lowers the number of women giving birth in the following year by approximately 3 per 1,000 rural registered-address women aged 20–49. This estimate remains negative across alternative definitions of fertility and precipitation exposure, regression specifications, and sample restrictions. The main mechanisms operate through lower agricultural income and greater labor migration, which respectively constrain resources for childbearing and disrupt household arrangements. We find no evidence of a health mechanism. Heterogeneity estimates provide no conclusive evidence that any measured geographic or economic characteristic attenuates the short-run fertility decline. Our findings therefore highlight fertility as an important margin of household adjustment to climate shocks and underscore the need to account for demographic responses when assessing the broader consequences of climate risk in rural economies.
Keywords: Extreme Precipitation; Fertility; Migration
JEL Codes: J13, O15, Q12, Q54

1 Introduction

Fertility theory emphasizes that changes in the resources and constraints surrounding childbearing can alter birth timing even without changing desired family size (Hirschman, 1994). Extreme precipitation can disrupt both. More than 85 percent of the population affected by dry rainfall shocks lives in low- and middle-income countries.[1] Rainfall and crop shocks reduce agricultural output (Chen et al., 2016), induce labor reallocation and migration (Minale, 2018), and adversely affect health (Dinkelman, 2017; Rocha and Soares, 2015). These effects are particularly salient in rural areas of developing countries, where livelihoods remain tied to climate-sensitive agriculture and formal insurance is limited. Yet the fertility response remains theoretically ambiguous. A realized income loss may induce couples to postpone childbirth (Alam and Pörtner, 2018), whereas persistent rainfall risk may increase fertility where children provide a buffer stock of family labor and formal insurance is limited (Cain, 1981; Grimm, 2021). Which force dominates depends on the livelihood and adjustment options available to households. This background motivates our examination of how fertility responds to extreme precipitation.

China provides an informative setting because substantial precipitation risk has coincided with pronounced changes in rural fertility and a rapid transformation of rural livelihoods. China's total fertility rate has fallen far below replacement level, with recent fertility declines particularly pronounced in rural areas.[2] At the same time, China has experienced marked regional changes in precipitation extremes.[3] These changes are particularly relevant to rural households because agriculture remained an important component of their livelihoods. Yet rural households increasingly supplemented farming with local nonfarm employment and self-employment (de Brauw et al., 2002), while rural–urban migration became an increasingly important source of employment and income (Meng, 2012). Rural China therefore provides a useful setting for examining how the fertility response to precipitation shocks varies as rural livelihoods and adjustment opportunities evolve.

We reconstruct county-year fertility histories for 2000–2014 from China's 2015 1% National

[1] *Ebb and Flow: Water, Migration, and Development*, derives this estimate from 189 population censuses in 64 countries, see https://www.worldbank.org/en/topic/water/publication/ebb-and-flow-water-migration-and-development. *The Unjust Climate* finds that floods reduce poor rural households' income by 4.4 percent relative to better-off households across 24 low- and middle-income countries, see https://www.fao.org/socioeconomic-research-analysis/resources/unjust-climate/the-unjust-climate/en.

[2] The World Bank reports China's total fertility rate at 1.01 births per woman in 2024, compared with a global average of 2.19; see https://data.worldbank.org/indicator/SP.DYN.TFRT.IN. China's 2020 Population Census reports a rural total fertility rate of approximately 1.54. Using the 2020 census to reassess fertility trends, Li et al. (2024) finds a wavelike decline in China's total fertility rate since 2010 and larger declines in rural than in urban areas across all seven regions between 2010 and 2020.

[3] *China's Fourth National Communication on Climate Change* reports that annual precipitation from heavy rainfall increased across most regions during 1961–2020, while meteorological drought worsened in Southwest China after the beginning of the twenty-first century.

Population Sample Survey and combine them with county-level precipitation records. The outcome is the number of women giving birth per 1,000 rural registered-address women aged 20–49. We measure local precipitation anomalies with the Standardized Precipitation Index (SPI) and classify a county-year as extreme when $|SPI| \geq 2$. Because the index locates rainfall within each county's historical distribution, it makes unusually dry and unusually wet conditions comparable across different precipitation regimes (Guttman, 1999). We estimate a county-year model that relates fertility to extreme precipitation in the preceding year and includes county fixed effects, province-by-year fixed effects, and lagged controls for temperature, specific humidity, and atmospheric pressure. We find that an extreme precipitation year is followed by 2.99 fewer women giving birth per 1,000 rural registered-address women. Relative to the sample mean of 36.12, this estimate corresponds to an 8.3% decline in next-year fertility. Estimates for the dry and wet tails are both negative and similar in magnitude. The relationship is nonlinear, fertility declines are concentrated in severe local anomalies rather than ordinary deviations.

We identify agricultural income and labor migration as two mechanisms underlying this fertility response. By using China Household Finance Survey data, we conclude that agricultural income losses reduce the resources available for childbearing, while migration changes employment and residential arrangements. The response among married adults further indicates that temporary spousal separation operates within the migration mechanism, consistent with evidence that migration and separation alter birth timing and birth intervals (Jin et al., 2019; Nie, 2020). By contrast, a broad measure of self-reported physical health shows no detectable change. The fertility relationship changes sign across periods. Using the 2000 National Population Census, we construct an additional county-level fertility panel for 1990–1999 within the same retrospective framework. Extreme precipitation is positively associated with fertility in the 1990s but negatively associated with fertility during 2000–2014. This contrast reflects a change in how rural households respond to agricultural risk. In a more agriculture-dependent setting, children have greater value as family labor and informal insurance (Cain, 1981; Grimm, 2021). As nonfarm work and migration expand, households gain other ways to diversify earnings, while agricultural income losses, employment transitions, and reduced co-residence raise the short-run cost of childbearing. The period comparison therefore shows how rural economic transformation changes the demographic response to similar precipitation anomalies.

The negative post-2000 estimate is stable across alternative reproductive-age ranges, fertility panels, mother-year specifications, precipitation measures, exposure windows and thresholds, richer controls and trends, clustering schemes, and geographic or disaster-related sample restrictions. The fertility response also does not differ significantly with local water infrastructure, agricultural mechanization, fiscal capacity, social networks, industrial structure, transport access, or different geography characters. The short-run decline is therefore widespread across the rural

settings measured in our data.

This paper contributes to the literature in three ways. First, we speak to the literature on climate shocks and demographic behavior by studying short-run fertility responses to locally extreme precipitation. Existing research has linked fertility to temperature shocks (Barreca et al., 2018; Sellers and Gray, 2019), hurricanes (Evans et al., 2010), water scarcity (Rocha and Soares, 2015), and agricultural income losses (Alam and Pörtner, 2018). More recent evidence from rural China documents lower inferred fertility following intense precipitation (Kou et al., 2025). We complement this work by defining exposure relative to each county's own historical precipitation distribution and examining exceptionally dry and exceptionally wet conditions within a common framework. This distinction is useful in a geographically diverse setting such as China, where the same absolute amount of rainfall can represent very different local conditions. Our results further show that the fertility response is concentrated in severe precipitation anomalies rather than moderate deviations from normal conditions. By measuring precipitation anomalies independently of recorded disaster status, the analysis includes severe dry and wet conditions regardless of whether they were formally classified as natural disasters.

Second, we connect the climate–fertility relationship to household labor reallocation during rural economic transformation. When weather shocks weaken local economic conditions, households may shift labor across sectors and locations (Minale, 2018). Migration can, in turn, alter the timing of childbearing and birth intervals (Jin et al., 2019; Nie, 2020). We examine these linked adjustment channels in a common empirical setting. Extreme precipitation is accompanied by lower agricultural income and greater migration among working-age adults, including married adults, but no detectable change in broad self-reported health. These responses place the fertility decline within a broader pattern of economic disruption and labor mobility and highlight migration as an important dimension of household adjustment to climate shocks.

Third, we add a development perspective by documenting opposite-signed, period-specific estimates during two phases of rural China's economic transformation. Existing studies using Chinese data have reported different fertility responses to precipitation shocks, including positive estimates for the 1990s (Tian et al., 2026) and negative estimates in later rural samples (Kou et al., 2025). Using an additional county-level fertility panel for 1990–1999, we similarly find positive estimates in the earlier period and negative estimates during 2000–2014. Over the same broad period, rural households gained greater access to nonfarm employment and rural–urban migration, expanding the margins available for responding to local shocks (de Brauw et al., 2002; Meng, 2012). The period comparison therefore places the climate–fertility relationship within an evolving rural economic environment and suggests that the demographic consequences of similar climate shocks can vary with households' livelihood and mobility options.

The remainder of the paper is organized as follows. Section 2 describes the institutional back-

ground, Section 3 introduces the data and variables, and Section 4 presents the empirical strategy and baseline results. Section 5 then examines accompanying evidence on income, migration, and health, followed by the heterogeneity analysis in Section 6. Section 7 concludes the empirical analysis with robustness checks, and Section 8 concludes.

2 Institutional Background

2.1 Fertility and Rural Demographic Change

Fertility has declined sharply worldwide, with a particularly pronounced decline in China. The global general fertility rate fell from 108.2 live births per 1,000 women aged 15–49 in 1990 to 66.5 in 2024, a decline of 38.5%. Over the same period, the corresponding rate in China fell from 89.4 to 27.9, a decline of 68.8%. Among Asian economies with data available in both years, China experienced the largest proportional decline.[4] China therefore entered our study period in the midst of a long-run fertility transition, although the pace and level of that transition differed markedly between rural and urban populations. In our retrospective fertility data, the rural registered-address sample has a higher fertility rate than the urban registered-address sample throughout most of 1990–2014, although both series decline over the period (Figure 1). The figure places the weather shocks studied in this paper within an ongoing transition toward lower fertility rather than a stable fertility regime.

Rural–urban differences in childbearing partly reflect differences in how households earn income, allocate labor, and support children. In rural China, childbearing was historically more closely connected to household production and the economic value of children's labor (Harrell et al., 2011), whereas fertility decline began earlier among urban and better-educated groups (Lavely and Freedman, 1990). This rural–urban divergence unfolded alongside changing fertility preferences and family aspirations (Whyte and Gu, 1987), broader socioeconomic and policy change (Cai, 2010), and China's transition from an agricultural to an increasingly urban and mobile society (Peng, 2011). Rural livelihoods, however, have become increasingly diversified. Farming has been complemented by local wage employment, self-employment, and rural–urban migration, so fertility decisions increasingly depend on a broader set of economic conditions that shape household income, labor allocation, and the time available for childrearing.

The fertility response to a local climate shock depends not only on the disruption itself, but also on the livelihood opportunities and adjustment margins available to rural households. This is especially relevant for the 2000–2014 period studied in our main analysis, when rural households increasingly combined farming with wage employment, self-employment, and mi-

[4] Author's calculations from United Nations, Department of Economic and Social Affairs, Population Division, *World Population Prospects 2024*, available at https://population.un.org/wpp/.

gration. As livelihood strategies diversified, the economic conditions surrounding childbearing also changed.[5] Children could still contribute labor and future support, but raising a child also required cash expenditure, caregiving time, and an expectation of stable future resources.

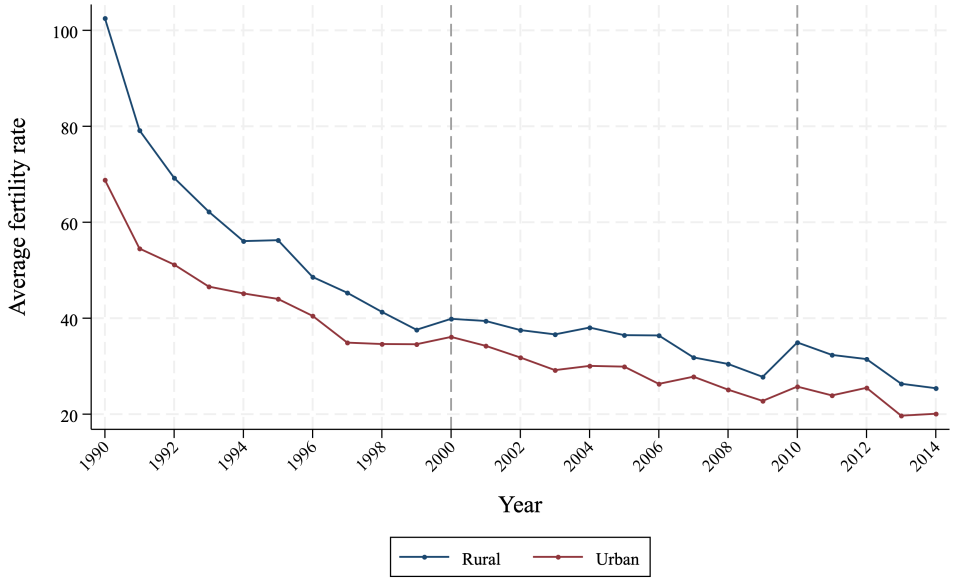


Fig. 1. Fertility Trends Across Urban and Rural Counties, 1990–2014

Notes: The figure uses the rural and urban retrospective fertility panels constructed from the 2015 1% National Population Sample Survey and plots the county-year mean fertility rate for each group. Fertility is the number of women giving birth per 1,000 registered-address women aged 20–49. The series are calculated after aggregating the individual county-year rates by calendar year, and the vertical reference lines mark 2000 and 2010.

2.2 Rural Livelihoods and Labor Mobility

Rural livelihoods in China became increasingly diversified over the period studied in this paper. Farming remained an important source of income and food, but rural households increasingly combined agricultural production with local wage employment, self-employment, and work outside the home county (de Brauw et al., 2002; Meng, 2012). This diversification gave households

[5]Direct official estimates of childrearing costs are not available for this period. We therefore use rural household consumption, together with expenditures in categories closely related to childrearing, as indirect evidence of changing household consumption commitments rather than direct measures of childrearing costs. According to the National Bureau of Statistics of China, rural per capita consumption expenditure increased from RMB 1,714 in 2000 to RMB 8,383 in 2014; see the NBS expenditure tables for 2000 and 2014 at https://www.stats.gov.cn/sj/ndsj/2008/html/J0925e.htm and https://www.stats.gov.cn/sj/zxfb/202302/t20230203_1898704.html. Based on the reported expenditure shares, per capita spending on education, culture, and recreation increased from approximately RMB 192 to RMB 863, while medical and health-care expenditure increased from approximately RMB 89 to RMB 754. Because the NBS household-survey framework changed during this period, comparisons across 2000 and 2014 should be read as descriptive. These measures therefore provide indirect evidence on changing household consumption commitments, not a direct estimate of childrearing costs.

additional ways to respond when local agricultural conditions deteriorated. Rather than absorbing an agricultural income loss entirely within farming, households could reallocate labor toward nonfarm activities or send working-age members elsewhere for employment. Road expansion further widened access to off-farm work (Qiao et al., 2014). Extreme precipitation may therefore affect rural households through two related margins: weather-sensitive agricultural income and the allocation of labor across sectors and locations.

Migration became an important part of this adjustment process. Rural–urban mobility expanded substantially from the 1990s onward, while the hukou system continued to separate registered residence from place of work and to limit migrants' access to urban schooling, housing, and social protection (Chan and Zhang, 1999; Song, 2014). Rural registration also remained connected to land, kin networks, and locally administered services. As a result, migration often involved only part of the household rather than permanent relocation of the entire family. A working-age adult could take a nonfarm job elsewhere while retaining land and family ties in the registered home county, allowing the household to diversify earnings without fully abandoning agricultural production.

This form of partial migration also reallocates labor and care within the household. A migrant's spouse, parents, or older children may remain in the origin county to farm, supervise children, or care for older relatives. Evidence from rural China documents changes in the time use of left-behind family members and consequences of parental migration for children's education and health (Chang et al., 2011; Meng and Yamauchi, 2017; Zhang et al., 2014). Remittances or other transfers may offset part of the income loss from local agricultural shocks, but they do not replace the time and labor of a resident adult. Existing evidence also shows that rural households respond to rainfall-induced agricultural losses through sectoral labor reallocation and temporary migration (Minale, 2018; Wang et al., 2024).

These adjustments have direct implications for fertility. Agricultural income losses tighten household budget constraints and may reduce the resources available for current and anticipated childrearing expenditures. Migration can buffer part of this loss by providing access to nonagricultural employment, but it can also reduce spousal co-residence, increase uncertainty over employment and residence, and shift caregiving responsibilities toward those who remain in the origin community. Evidence from China shows that migration-related spousal separation affects birth timing and birth intervals (Jin et al., 2019; Nie, 2020), with related evidence from other migration settings pointing to similar short-run disruptions (Clifford, 2009). Labor reallocation can therefore both smooth household income and alter the economic and residential conditions surrounding childbearing. We accordingly examine agricultural income as a measure of weather-sensitive household resources and working-age migration as an observable margin of labor-market adjustment following extreme precipitation.

2.3 Extreme Precipitation and Adaptation

China's precipitation environment is spatially and seasonally heterogeneous. The monsoon delivers much of the annual rainfall during the growing season, while counties differ sharply in their normal precipitation levels, the timing of rainfall, and exposure to dry spells or excessive rain. A national review likewise emphasizes strong regional contrasts in observed precipitation and water-resource trends and substantial uncertainty about their net consequences for agriculture (Piao et al., 2010). The same amount of rainfall therefore carries different historical and economic significance across counties. The analysis accordingly defines exposure using each county's own historical precipitation distribution. Appendix A.2 describes this construction, and Appendix Figure A2 maps changes from 2000 to 2014 in dry days and days above the county-specific historical 95th percentile of wet-day precipitation.

Extreme precipitation can disrupt rural economic activity even when it does not result in a formally recorded natural disaster. Unusually dry or wet conditions may reduce agricultural production, damage assets, interrupt work, and induce household adjustment without necessarily causing immediate injury, mortality, or visible physical destruction. By contrast, floods are realized outcomes that depend not only on the underlying precipitation shock but also on local topography, drainage and river systems, and protective infrastructure. The same precipitation anomaly may therefore have different realized consequences across locations, while still affecting agricultural production and household livelihoods. This distinction motivates our focus on the underlying precipitation anomaly rather than on recorded disasters alone. It is also important to consider both tails of the precipitation distribution, because dry and wet extremes operate through different physical channels. Drought constrains water availability and crop growth, whereas excessive rainfall can delay planting and harvesting, damage crops, and overwhelm local drainage systems.[6]

The economic consequences of a precipitation anomaly depend not only on the severity of the shock but also on the capacity of local production systems and households to adjust. Water-control infrastructure can directly moderate agricultural exposure. Reservoirs and irrigation systems can reduce water shortages during dry periods, while drainage and flood-control facilities can limit losses from excessive rainfall. Farmers may also adjust production decisions after a shock by changing crop choice, shifting planting or harvesting dates, reallocating land across crops, modifying fertilizer and other input use, or relying more heavily on machinery and hired labor. These responses can reduce yield losses or allow production to recover more quickly,

[6]Evidence on rainfall, agricultural production, and local economic activity documents nonlinear effects at both tails (Chen et al., 2016; Damania et al., 2020). International assessments similarly distinguish meteorological hazards from their realized social consequences, which depend on exposure and vulnerability. See the Intergovernmental Panel on Climate Change, *Climate Change 2022: Impacts, Adaptation and Vulnerability*, available at https://www.ipcc.ch/report/ar6/wg2/.

although their effectiveness depends on access to capital, information, infrastructure, and local markets. Beyond the farm, households can further buffer income losses by drawing on savings or informal support, reallocating labor toward nonagricultural employment, or sending household members elsewhere for work.[7] The same precipitation anomaly may therefore generate different economic losses across counties depending on the availability of these adjustment margins. This motivates the heterogeneity analysis below, which examines whether the fertility response varies with pre-existing water-control infrastructure, agricultural mechanization, fiscal capacity, informal support, industrial structure, and transport access.

3 Data and Variables

3.1 Fertility Data and Variables

We use the 2015 1% National Population Sample Survey to reconstruct county-year fertility histories for 2000–2014. Because the survey does not provide a direct mother–child identifier, we infer maternal links from household relationships and recover childbirth events from children's reported years and months of birth, building on the household-roster approach of Tian et al. (2026). We limit the retrospective window to children born from 2000 onward to reduce the omission of older children who may have left the household before the 2015 survey. For each year from 2000 to 2014, we calculate each woman's age in that year from her reported year of birth and construct the corresponding population of women aged 20–49. Thus, the age-eligible sample is defined separately in each year rather than by women's ages at the time of the 2015 survey. We retain women with rural registered addresses and assign weather exposure using the county of registration. To reduce exposure misclassification, the baseline sample excludes women who report prolonged absence from their registered county. Appendix A.1 describes the rural classification, absence restriction, mother–child linkage, age reconstruction, and aggregation procedure. We aggregate inferred childbirth events to the mother's registered-address county and the child's year of birth, counting each woman once in a year in which she gives birth.

Following the demographic convention of expressing fertility relative to women of reproductive age, we define the main outcome as the number of rural registered-address women giving birth per 1,000 women in the same county-year population.[8] Each woman is counted once for

[7] Examples of these adjustment margins include irrigation and water infrastructure (Blakeslee et al., 2023; Huang and Dong, 2025), agricultural input and production responses (Chen and Gong, 2021), consumption smoothing and crop-portfolio adjustment (Cui and Tang, 2024), limits to adaptation (Burke and Emerick, 2016), and labor reallocation or informal risk sharing (Mazur, 2023; Minale, 2018). The heat-focused studies provide evidence on adjustment margins across climate hazards rather than direct estimates of adaptation to the precipitation measure used here.

[8] International demographic metadata conventionally use ages 15–49 to define women of reproductive age; see the World Health Organization's fertility-rate metadata at https://www.who.int/data/gho/

a delivery, irrespective of the number of children born. The baseline analysis uses women aged 20–49 and reconstructs fertility for 2000–2014. Restricting the retrospective window limits the omission of older children (aged 15 and above) who may already have left the household, while excluding 2015 avoids treating the incomplete survey year as a full calendar year. In the robustness analysis, we extend the age range to 15–49 and construct an alternative fertility panel that combines the 2010 Population Census with the 2015 survey.

3.2 Weather Variables

We obtain precipitation and other weather variables from a county-level panel constructed from the Global Surface Summary of the Day (GSOD) database maintained by the NOAA National Centers for Environmental Information. [9]

We summarize annual precipitation using a county-specific Standardized Precipitation Index (SPI), following the established climatological literature (McKee et al., 1993). We fit each county's 1973–1997 precipitation distribution and transform the resulting cumulative probability to the standard-normal scale to construct SPI. Negative values indicate relatively dry conditions, while positive values indicate relatively wet conditions. Larger absolute values correspond to more severe departures from the local historical norm.[10] Because SPI can be calculated over different accumulation horizons, it is widely used to characterize both the severity and duration of moisture anomalies (Guttman, 1999). Applications include drought monitoring in the United States (Hayes et al., 1999), assessments of antecedent wetness and flood risk (Seiler et al., 2002), analyses of drought intensity and spatial extent (Edossa et al., 2010; Rouault and Richard, 2003), and studies of both dry and wet conditions in Europe and China (Bordi et al., 2009; Wang et al., 2022). Appendix Figure A1 maps the average county-level SPI during 2000–2014.

Our main exposure is an extreme-precipitation indicator (EP) that equals one when the absolute value of a county's annual SPI reaches or exceeds 2, that is, when SPI is at or below -2 or at or above 2. The two-sided definition reflects evidence that rainfall has nonlinear effects on agricultural production and local economic activity. Both unusually dry and unusually wet conditions can therefore disrupt rural livelihoods (Chen et al., 2016; Damania et al., 2020). We relate EP to

indicator-metadata-registry/imr-details/123. Our baseline measure uses women aged 20–49 because births below age 20 are relatively uncommon in the study population; the broader 15–49 definition is examined as a robustness check.

[9]The raw GSOD product provides daily summaries from land-based weather stations. The county-level panel spatially interpolates station observations to a $0.1^\circ \times 0.1^\circ$ grid using the five nearest stations, converts the interpolated grid to raster format, and averages raster cells within county boundaries. We subsequently harmonize county identifiers and boundaries to the 2014 administrative divisions used throughout the analysis. Documentation for the underlying GSOD product is available at https://www.ncei.noaa.gov/access/metadata/landing-page/bin/iso?id=gov.noaa.ncdc:C00516.

[10]See World Meteorological Organization, 2012, *Standardized Precipitation Index User Guide*, available at https://library.wmo.int/zh/records/item/39629-standardized-precipitation-index-user-guide.

fertility in the following year, consistent with the timing of conception and pregnancy before the observed birth. We assess robustness to alternative SPI thresholds, replace annual exposure with a z-score measure, and use standardized annual precipitation in place of SPI. We also estimate continuous SPI together with its square as a functional-form check, allowing the fertility response to vary nonlinearly across the precipitation distribution. Appendix A.2 provides the full SPI construction, alternative threshold definitions, temporal alignment, and administrative-boundary harmonization.

The baseline regressions additionally control for temperature, specific humidity, and atmospheric pressure from the same county-level weather panel. Temperature is included because crop yields and agricultural productivity in China respond to thermal conditions (Chen et al., 2016; Chen and Gong, 2021). Specific humidity measures the mass of water vapor per unit mass of moist air and therefore captures atmospheric moisture without scaling by the temperature-dependent saturation level, as relative humidity does. This distinction can matter empirically because accounting for humidity changes estimated climate effects on Chinese crop yields (Zhang et al., 2017). Atmospheric pressure provides an additional control for broader synoptic weather conditions that co-vary with precipitation. Previous county-level studies in China similarly include pressure among ancillary weather controls (Li et al., 2020). All three controls are lagged by one year to match the timing of the precipitation exposure and standardized within county using their 1973–1997 historical means and standard deviations. They therefore capture local weather anomalies on the same temporal and historical reference basis as the precipitation measure.

3.3 Household and Individual Variables

We use the China Household Finance Survey (CHFS) to examine household-level responses that are not observed in the population survey. The CHFS is a large-scale biennial household survey that collects detailed information on household finances, employment, migration, health, and household composition (Gan et al., 2016). We use the 2011, 2013, and 2015 waves to construct three sets of accompanying outcomes: household income, migration, and self-reported health. We measure household resources using total household income and agricultural operating income for households whose head is younger than 60. We measure migration among working-age adults aged 16–59 and classify an individual as a migrant when he or she was away from home for more than six months during the relevant survey period. We also construct a binary measure of self-reported physical health from the original five-category scale and examine it across the relevant age and gender groups.

We align weather exposure to the reference period of each CHFS outcome. The income measures refer to the preceding calendar year, so the 2011, 2013, and 2015 waves provide income outcomes for 2010, 2012, and 2014. The migration and health outcomes are likewise assigned to

2010, 2012, and 2014 according to their survey reference periods and matched to precipitation in the same calendar year. This timing allows the CHFS regressions to capture economic, mobility, and health responses associated with the precipitation shock before the fertility outcome observed in the following year.

3.4 County-Level Variables

The county-level adaptation analysis combines agricultural and economic indicators with spatial infrastructure data. Agricultural output, machinery, rural income, labor, and industry measures come from the county statistical yearbook series. Dam locations come from the Global Reservoir and Dam Database (GRanD) (Lehner et al., 2011).[11] Related evidence examines whether flood-control dams moderate the educational effects of flooding (Huang and Dong, 2025). We classify a county as covered by water-control infrastructure when a GRanD-recorded irrigation dam is located within the county or in a contiguous county. Appendix A.3 details this spatial extension and the remaining county-level variables.

3.5 Summary Statistics

Table 1 reports summary statistics for the estimation samples and variables used in the main analysis, mechanism analysis, and heterogeneity analysis. Panel A shows that the mean fertility rate is 36.12 women giving birth per 1,000 rural registered-address women, compared with 29.08 in the corresponding urban population. This difference is consistent with the higher rural fertility series shown in Figure 1. Panel B shows that extreme precipitation occurs in about 5.2% of county-years in the baseline sample, indicating that the preferred SPI threshold captures relatively infrequent local precipitation anomalies.

Panel C summarizes the county characteristics used in the heterogeneity analysis. About 57% of county-years have a nearby dam. The average primary-industry share is 0.25, compared with 0.42 for the secondary-industry share, while average motorway density is 30.64 metres per square kilometre. Observation counts are lower for several county-level variables because the underlying local statistical yearbooks contain substantial missing information for some counties and years.

Panel D describes the rural household and individual samples used in the mechanism analysis. Household heads are predominantly male and married, with an average age of 46.7 years and about 8 years of schooling. The average total household income is approximately 45.42 thousand yuan. Approximately 25% of individuals in the migration sample are classified as migrants, while 67% of respondents in the health sample report better self-rated health. Differences in observa-

[11]The GRanD inventory is maintained by Global Dam Watch. Data and documentation are available at https://www.globaldamwatch.org/grand/.

tion counts across these outcomes reflect the outcome-specific samples used in the mechanism analysis.

[Insert Table 1 about here]

4 Empirical Strategy and Results

4.1 Empirical Strategy

We estimate the relationship between lagged extreme precipitation and fertility using the following county-year fixed-effects model, which relates fertility in year t to extreme precipitation in year $t-1$ and a vector of lagged controls:

$$Fertility_{ct} = EP_{c,t-1}\beta + X_{c,t-1}\gamma + \alpha_c + \lambda_{pt} + \epsilon_{ct}, \tag{1}$$

where c indexes counties, p indexes provinces, and t denotes the year of the fertility outcome. The dependent variable, $Fertility_{ct}$, measures the number of rural registered-address women aged 20–49 who give birth per 1,000 women in county c and year t. Each woman is counted once per delivery. The main exposure, $EP_{c,t-1}$, equals one when $|SPI_{c,t-1}| \geq 2$ in the preceding year, capturing either an extremely dry or an extremely wet condition. We assign weather exposure in year $t-1$ to fertility in year t because conception and pregnancy precede the observed birth. With annual data, this lag provides an approximation to the relevant preconception and gestational exposure period.

The vector $X_{c,t-1}$ includes lagged temperature, specific humidity, and atmospheric pressure, each standardized relative to the county-specific 1973–1997 reference period. County fixed effects, α_c, absorb time-invariant county characteristics, while province-by-year fixed effects, λ_{pt}, absorb province-specific annual conditions, including policy, public-service, and economic changes shared by counties within the same province. The coefficient β is identified by within-county changes in extreme precipitation relative to other counties in the same province and year, conditional on the lagged climate controls. Because the estimate may remain sensitive to model specification, omitted time-varying local factors, and alternative definitions of exposure or fertility, we examine robustness to alternative weather measures, fertility outcomes, exposure windows, time trends, additional controls, clustering levels, and sample restrictions. Standard errors in the baseline specification are clustered at the county level to allow for heteroskedasticity and serial correlation within counties over time.

4.2 Baseline Results

We first examine the relationship between the occurrence of extreme rainfall and fertility in the subsequent year. Table 2 reports the baseline estimates. Columns (1)–(3) include county and year fixed effects and examine the extremely dry, extremely wet separately and jointly. We define an extremely dry year as one in which $SPI \leq -2$, and an extremely wet year as one in which $SPI \geq 2$, and zero otherwise. In column (3), the coefficients are -3.132 for extremely dry conditions and -3.515 for extremely wet conditions. Both estimates are negative, precisely estimated, and similar in magnitude. Column (4) combines the two tails into a single extreme-precipitation indicator, defined as one when $|SPI| \geq 2$ and zero otherwise. Column (5), our preferred specification, replaces year fixed effects with province-by-year fixed effects. The estimated coefficient is -2.992, indicating that an extreme precipitation event is followed by 2.99 fewer women giving birth per 1,000 rural registered-address women aged 20–49 in the following year. Relative to the baseline mean of 36.117 women per 1,000, this corresponds to an 8.3% decline.[12] The baseline estimate pertains to severe local precipitation anomalies rather than rainfall generally. Section 7 therefore examines alternative thresholds and a quadratic SPI specification to assess how the estimated relationship varies across the precipitation distribution.

Table 2 shows that fertility falls after both extremely dry and extremely wet years. Columns (1)–(3) include county and year fixed effects and estimate the two tails separately and jointly. When both indicators enter together in column (3), the coefficient is -3.132 for an extremely dry year and -3.515 for an extremely wet year, with standard errors of 1.155 and 1.196. Both estimates are significant at the 1% level and are similar in magnitude.

[Insert Table 2 about here]

5 Mechanism Analysis

To clarify the findings in baseline estimates, we examine three potential channels: household income, migration, and physical health. First, household income may affect fertility by shaping the resources available for childbearing. Income losses can constrain current expenditure on a child and weaken expectations that parents can sustain the economic and care inputs required after birth, leading households to postpone childbearing. Second, migration may affect fertility

[12] A related estimate for rural China is reported by Kou et al. (2025), who find that a one-percentage-point increase in the annual share of days with intense precipitation reduces the following year's number of births per household by 6.37×10^{-3}. Their treatment is a continuous measure of intense-rainfall frequency, whereas ours identifies county-years in which the annual SPI crosses either the -2 or 2 threshold. Their outcome is an inferred household-level birth count, whereas ours measures the number of women giving birth per 1,000 women. Despite these differences in treatment and outcome definitions, both studies document a negative fertility response to extreme precipitation, and the implied magnitudes are broadly comparable when expressed at a common regional scale.

through changes in co-residence and family formation. For example, living apart may reduce opportunities for conception and be associated with longer corresponding birth intervals. Third, physical health can also influence couple's conception, pregnancy, and childbirth. We use the 2011, 2013, and 2015 CHFS waves to supply some household evidences to test these potential channels.

5.1 Agricultural Income

Agricultural income provides the first link between extreme precipitation and fertility. Unusually dry or wet conditions can reduce agricultural production and increase income uncertainty, tightening household resources and making the immediate costs of childbearing and the subsequent demands on income and caregiving time more difficult to absorb, thereby inducing couples to postpone childbearing. Evidence from Tanzania shows that crop losses reduce pregnancies and births while increasing contraceptive use, consistent with fertility postponement after agricultural income shocks (Alam and Pörtner, 2018). Recent evidence from rural China likewise links intense precipitation to lower inferred fertility and presents seasonal patterns consistent with an income-expectation mechanism (Kou et al., 2025). Households may respond through nonfarm employment, yet these adjustments can carry their own costs for childbearing.

Table 3 shows the response of household incomes. Columns (1) and (3) are indicators for whether the household reports any total or agricultural income, respectively. Columns (2) and (4) use the corresponding inverse-hyperbolic-sine (IHS) measures.[13] The income measures refer to the previous year, so the dependent variables cover 2010, 2012, and 2014 and are matched to extreme precipitation in the corresponding year. All specifications include household and climate controls, county fixed effects, and province-by-year fixed effects.

Neither total-income outcome shows a detectable response. By contrast, EP reduces the probability that a household reports positive agricultural income by 9.2 percentage points. The corresponding coefficient for IHS-transformed agricultural income is -0.897, capturing the combined response along the extensive and intensive margins. The estimates are consistent with adjustment across income sources because total income combines agricultural earnings, nonfarm earnings, and transfers, whereas agricultural income directly captures the weather-sensitive component of the household budget. The agricultural-income estimates provide accompanying evidence

[13]We retain household income outcomes for households whose head is younger than 60. Income measures reported in levels are transformed using the inverse hyperbolic sine, which accommodates zero and small values while reducing the influence of extreme observations. We use the inverse hyperbolic sine transformation, $\operatorname{asinh}(x) = \ln\left(x + \sqrt{x^2+1}\right)$, which is defined at zero and for negative values (Burbidge et al., 1988). As emphasized by Chen and Roth (2024), treatment effects on log-like transformations can depend on the units of measurement when treatment changes whether an outcome is zero. We therefore report these coefficients in transformed-outcome units.

of a weather-sensitive financial disruption at the household level. Related evidence from (Wang et al., 2024), which links extreme rainfall to lower agricultural yields and income and to labor reallocation across sectors. To place this household evidence in a county-level context, Appendix Table B4 reports agricultural and industrial outcomes.[14] Panel A shows no detectable contemporaneous change in county agricultural GDP or its sectoral share. This muted aggregate response need not conflict with the decline in household agricultural income. Rural production in China is dominated by smallholders, whose capacity to absorb and adapt to climate shocks may differ substantially from that of larger producers. Household agricultural income may therefore respond unevenly across producers even when county-level agricultural output changes little. Panels B and C provide complementary evidence on subsequent adjustment. Cotton output increases, as does the secondary-industry share, whereas the estimates for oil-crop output, industrial GDP, and average rural income show no detectable changes. Together, these results point to adjustments in crop allocation and sectoral composition alongside the household-level agricultural income response.

Taken together, the household and county-level estimates are consistent with an agricultural-resource channel. The household evidence identifies the clearest response in agricultural income, while the county-level results suggest adjustments in the composition of local production and economic activity rather than a broad contraction in aggregate output. This provides the economic context for the migration response examined next.

[Insert Table 3 about here]

5.2 Migration and Spousal Separation

Migration is an important adjustment margin when agricultural shocks reduce local earnings. Working-age adults may seek employment in less affected locations or nonagricultural sectors, thereby diversifying household income and spreading weather risk across places. Existing evidence documents climate-induced labor reallocation and migration across settings and shows that transport access can facilitate nonagricultural employment.[15] The resulting sustained absence may affect fertility by reducing spousal co-residence or altering matching conditions in local marriage markets.

[14] Appendix B.3 discusses these estimates in detail.

[15] Rural Chinese households have been shown to respond to rainfall-induced agricultural losses through sectoral labor reallocation and temporary migration (Minale, 2018; Wang et al., 2024). Evidence from Brazil similarly shows that persistent drying reduces agricultural employment and induces worker reallocation and climate migration, while financial and spatial frictions constrain adjustment (Albert et al., 2021). Beyond predominantly agrarian settings, US evidence finds that warmer temperatures induce net out-migration and that cumulative responses grow over longer horizons (Baylis et al., 2025). Road expansion also facilitates local off-farm employment in rural China (Qiao et al., 2014).

We follow the CHFS migration definition and classify an individual as a migrant when they were away from home for more than six months during the survey year.[16] The resulting duration-based measure captures sustained absence. Using the survey's timing information, we match the migration outcome to the corresponding weather year. Table 4 examines this outcome among adults aged 16–59. We estimate a 3.2-percentage-point increase in migration in the full working-age sample. The point estimates are 3.9 percentage points for men and 2.6 percentage points for women, with the latter statistically imprecise. Columns (4)–(6) show a larger response among unmarried adults: migration increases by 5.9 percentage points for the full unmarried sample and by 8.1 percentage points for unmarried men, while the estimate for unmarried women is statistically imprecise. We report the migration response among married working-age adults in Appendix Table B5, where the estimated increase is 2.7 percentage points.

The migration response among married adults is consistent with temporary spousal separation as one potential fertility channel. Reduced co-residence may lower opportunities for conception during periods of separation, while employment and residential uncertainty may lead couples to postpone childbearing.[17] Among unmarried adults, the comparatively large male point estimate is also consistent with male out-migration increasing matching frictions in origin marriage markets and thereby delaying marriage and subsequent childbearing, although the absence of cross-gender equality tests limits this interpretation. Fertility may additionally be postponed after migration as individuals adjust to employment, housing, and hukou-related constraints at their destinations.[18] Our migration measure does not distinguish these origin- and destination-side pathways. The estimates therefore provide accompanying evidence for migration as a relevant adjustment margin, rather than direct evidence that spousal separation or destination adjustment mediates the fertility response.

[Insert Table 4 about here]

5.3 Physical Health

Severe weather can increase exposure to injury, infectious disease, disruptions in medical care, and nutritional stress, which may in turn affect conception, pregnancy, and childbirth. Existing evidence documents lasting effects of rainfall conditions at birth (Maccini and Yang, 2009)

[16]The migration indicator equals one when an individual was away from home for more than six months during the survey year. The main analysis focuses on working-age individuals aged 16–59. In the CHFS, individuals who have been away for more than six months remain members of the surveyed household if they have not become economically independent. The measure therefore generally captures temporary labor migration by individuals who retain economic ties to, and are expected to return to, their origin household rather than permanent departure to establish a separate household.

[17]Previous studies find that migration-related spousal separation can disrupt fertility in the short run (Clifford, 2009; Nie, 2020).

[18]Using the 2017 China Fertility Survey, Jin et al. (2019) document longer birth intervals among women who migrated between marriage and their first birth or between their first and second births.

and adverse fetal exposures (Almond and Currie, 2011).[19] We use the CHFS self-reported health measure to test whether extreme precipitation is associated with changes in physical health. [20] Table 5 reports estimates for adults aged 16 or above, adults aged 16–49, women aged 16–49, and adults aged 60 or above. We match the 2010, 2012, and 2014 health outcomes to extreme precipitation in the corresponding year.

Across all four samples, we estimate negative coefficients that are statistically indistinguishable from zero, including among women of reproductive age. We therefore find no detectable change in broad self-reported physical health following extreme precipitation. Taken together, the mechanism analysis points primarily to agricultural-income disruption and labor mobility, while providing no detectable evidence for the broad self-reported health mechanism measured here.

[Insert Table 5 about here]

6 Heterogeneity

6.1 Adaptation Capacity

We next examine whether the fertility response varies with local capacity to absorb agricultural-income shocks. From the perspective of agricultural adaptation, irrigation can protect production against drought by reducing dependence on natural rainfall, while agricultural modernization can make crop production less sensitive to weather conditions. We capture these dimensions using nearby reservoir presence, the local level of agricultural mechanization, and government fiscal expenditure. Reservoirs proxy access to irrigation infrastructure, mechanization captures the extent of agricultural modernization, and greater fiscal resources may allow local governments to provide more support following adverse shocks. We use these measures to examine whether local adaptive capacity can cushion rural households against the fertility consequences of extreme precipitation.

Columns (1)–(3) of Table 6 show that the interactions with reservoir presence and mechanization are positive, whereas the interaction with fiscal capacity is negative. None of these

[19]These findings motivate our examination of health alongside the income and migration channels. Studies of water scarcity and hurricane exposure during pregnancy also find effects on infant survival and health at birth (Currie and Rossin-Slater, 2013; Rocha and Soares, 2015). Related evidence links drought exposure in infancy to later-life disability in South Africa (Dinkelman, 2017) and shows that low growing-season rainfall increases infant mortality in Uganda when access to community health workers is limited (Björkman Nyqvist et al., 2025). Disaster-related mortality may also affect subsequent childbearing through responses to child loss and community mortality (Nobles et al., 2015).

[20]The health outcome is binary. We code responses 1–2 on the original five-category self-reported health scale as zero and responses 3–5 as one, so a larger value denotes better health.

interaction estimates is statistically distinguishable from zero, although several are economically nontrivial in magnitude. In particular, the positive coefficients for reservoir presence and mechanization, together with the negative coefficient for fiscal capacity, suggest the possibility that tangible adaptation measures that directly reduce households' exposure to climate risk may be more effective than broader fiscal capacity in strengthening households' resilience to extreme precipitation.

Fertility decisions may also be shaped by local clan institutions. Clan networks can reinforce preferences for lineage continuation and provide social capital that helps families share the costs of childbearing (Cao et al., 2022). Areas with stronger clan traditions may therefore display both stronger preferences for childbearing and greater informal support for families facing adverse income shocks. We use genealogy density as a proxy for the strength of local clan networks and examine whether the fertility response to extreme precipitation varies with this cultural and social environment. In column (4) of Table 6, the interaction with genealogy density is positive, indicating a smaller fertility decline in places with denser genealogical networks.

[Insert Table 6 about here]

6.2 Employment and Mobility Conditions

Industrial structure and transport access may shape how rural households adjust when agricultural returns fall. Counties with a larger secondary sector may provide greater access to nearby nonfarm employment, whereas counties that depend more heavily on the primary sector may offer fewer alternatives to weather-sensitive agriculture. Nonfarm activity, however, is not fully insulated from weather shocks. Evidence from Chinese manufacturing shows that extreme heavy-rainfall anomalies reduce productivity through disruptions to labor, agricultural intermediate inputs, and transport (Chen et al., 2025). These patterns are consistent with the broader expansion of nonfarm employment and labor reallocation in rural China (de Brauw et al., 2002; Meng, 2012; Minale, 2018).

Table 7 reports interaction models based on above-median primary- and secondary-industry shares and pre-existing motorway access in 2000. The interaction with a larger primary-industry share is negative, whereas that with a larger secondary-industry share is positive, although neither estimate is statistically significant. Column (3) shows that EP is associated with 3.046 fewer women giving birth per 1,000 women in counties without pre-existing motorway access. The interaction with motorway access is positive at 1.092, indicating a smaller fertility decline in counties with motorway access, but the difference is not statistically significant. Appendix Table B3 provides complementary evidence across household employment structures. The estimated effects are negative across agricultural, mixed-employment, and nonagricultural households, with

the decline being statistically significant only for agricultural households. Sum up, the signs of these estimates are consistent with the idea that access to nonfarm employment and lower mobility costs may broaden households' adjustment options following an agricultural shock, although the estimated differences across groups are not statistically significant.

[Insert Table 7 about here]

6.3 Economic Transition and the Fertility Response

The fertility effects of agricultural and climate risk vary with households' economic environment and capacity to absorb shocks. In settings with limited access to capital markets, public relief, or other forms of insurance, children may provide greater economic value as a source of labor and future support (Cain, 1981). Consistent with this view, rainfall risk has been associated with larger fertility differentials between farm and nonfarm households, while irrigation and agricultural machinery weaken this relationship (Grimm, 2021). Evidence on realized climate shocks is similarly heterogeneous across settings. In Bangladesh, flood exposure increases childbearing among less-educated and higher-parity women but reduces it among childless women and urban residents (Thiede et al., 2025). Other studies document lower fertility following adverse agricultural conditions, including sustained heat in Indonesia (Sellers and Gray, 2019) and crop losses in Tanzania (Alam and Pörtner, 2018).

We therefore compare our post-2000 estimates with an earlier period. Following Tian et al. (2026), we reconstruct a county-level fertility panel for 1990–1999 from the 2000 National Population Census of China.[21] Table 8 reports positive estimates for the 1990s and a negative estimate for 2000–2014. The comparison can be interpreted in light of differences in rural households' adjustment opportunities. Off-farm employment and migration have become important margins of labor allocation in rural China (de Brauw et al., 2002; Meng, 2012). More directly, rural households respond to negative agricultural productivity shocks by reallocating labor away from farming and toward migration and off-farm work (Minale, 2018). Road expansion can further expand local nonfarm opportunities by increasing participation, working time, and income in off-farm employment (Qiao et al., 2014). At the same time, the household-registration system continued to shape the costs of rural–urban mobility (Chan and Zhang, 1999; Song, 2014). These changes may reduce reliance on children as a source of labor or insurance, while migration and employment transitions may increase uncertainty and disrupt spousal co-residence.

[21] In table 8, columns (1)–(2) use county and year fixed effects and SPI thresholds of 1.5 and 2.0 in our reconstructed 1990s panel. Tian et al. (2026) use monthly births per 100 women aged 15–45, align weather with conception using a 10-month birth lag, and include county, year, and month fixed effects. Their SPI uses Chinese station records over 1951–2000, whereas ours uses GSOD data and a 1973–1997 reference period. Our annual outcome counts women giving birth per 1,000 women. Consequently, columns (1)–(2) provide a period comparison under our retrospective design. Column (3) reports the 2000–2014 baseline result using county and province-by-year fixed effects.

[Insert Table 8 about here]

6.4 Geographic and Agroclimatic Checks

Previous evidence suggests that climate impacts can vary systematically across geographic and terrain conditions. The Qinling–Huaihe Line provides a conventional north–south division across major temperature and moisture regimes in eastern China. More generally, China's climate regions differ in thermal resources, aridity, and precipitation, and topography is itself an important determinant of local climatic conditions and climatic boundaries (Zheng et al., 2013). These differences shape growing seasons, water availability, and the agricultural consequences of rainfall shocks.Regionally differentiated effects of climate change on water resources and agriculture (Piao et al., 2010). Rural households' income and labor allocation depend on these local production conditions, while evidence from agricultural income shocks indicates that household resources can affect the timing of childbearing (Alam and Pörtner, 2018). In our setting, these two links imply that an extreme precipitation shock may generate different fertility responses across the north–south divide and terrain gradients. We therefore test whether the fertility response differs by these two geographic characteristics.

Appendix B.5 Table B6 reports pooled interactions of EP with a North indicator and an above-median terrain-slope indicator. The interaction coefficients are positive for both northern counties and counties with steeper terrain, but neither is statistically significant. The estimates therefore provide no evidence that the fertility response differs systematically across the north–south divide or by terrain slope. Overall, the negative fertility response appears broadly similar across these geographic and agroclimatic conditions.

7 Robustness Analysis

7.1 Alternative Measures

The baseline uses SPI because it measures precipitation relative to a county's own historical distribution. The first set of checks evaluates sensitivity to the measure and threshold used to define severe local precipitation conditions. Table 9 shows a negative estimate when EP is constructed from a conventional annual rainfall z-score, while the estimates become more negative as the SPI threshold becomes more stringent. The coefficient at $|SPI| \geq 2.5$ is -4.320, compared with -1.024 at the 1.5 threshold, where the estimate is not statistically significant. Figure 2 reports the estimated coefficients across different SPI thresholds. The alternative standardization and threshold estimates therefore indicate that the negative relationship is concentrated in severe local precipitation anomalies rather than broader deviations from normal conditions.

Column (4) instead enters continuous SPI and its square. The negative quadratic term is compatible with fertility declining in both tails of the local precipitation distribution, a pattern that parallels the nonlinear precipitation relationships documented for agricultural production and economic activity (Chen et al., 2016; Damania et al., 2020). We retain $|SPI| \geq 2$ as the preferred definition because it corresponds to the conventional extremely dry and extremely wet SPI categories (Hayes et al., 1999; McKee et al., 1993). In our sample, this threshold also retains sufficient extreme-event observations for estimation. Appendix Figure B1 shows an inverted-U relationship between fertility and precipitation conditions, with fertility declining toward both tails of the distribution. To provide a more intuitive interpretation of these thresholds, we also define extreme precipitation using alternative upper and lower percentiles of the historical precipitation distribution. By mapping percentile cutoffs to the corresponding SPI values, Appendix Table B2 reports the results. The coefficient pattern is consistent with the main finding that the fertility response is concentrated in the two tails of the precipitation distribution. [22]

[Insert Table 9 about here]

[Insert Figure 2 about here]

7.2 Alternative Outcomes and Exposure Windows

Table 10 examines whether the baseline relationship is sensitive to alternative outcome definitions, precipitation windows, and retrospective data sources. Column (1) broadens the reproductive-age denominator from women aged 20–49 to the conventional range of 15–49. The estimated coefficient remains nearly unchanged at -3.079. Column (2) moves from the county-year fertility rate to a woman-year panel in which the dependent variable indicates whether a woman gives birth in a given year. The estimate remains negative and statistically significant. Its magnitude, -0.002, corresponds to approximately two fewer women giving birth per 1,000 women, and is therefore comparable in direction and scale to the baseline result.

Column (3) replaces annual EP with an indicator constructed from March–September growing-season precipitation.[23] The coefficient remains negative, indicating that the result is not specific to the annual precipitation measure. Finally, column (4) combines fertility histories reconstructed from the 2010 National Population Census for 2000–2009 with those from the 2015 survey for 2010–2014. The estimate is again negative and statistically significant, reducing concern that the baseline result is driven by retrospective measurement in a single survey

[22] SPI uses a fitted county-specific precipitation distribution, whereas the percentile check classifies annual precipitation directly using the 1973–1997 empirical distribution. Appendix A.2 gives the precise construction and correspondence between the two measures.

[23] Following Björkman Nyqvist et al. (2025), we define the main agricultural growing season as March through September of each year.

source. Overall, the negative fertility response persists across alternative age definitions, levels of aggregation, exposure windows, and fertility panels. Appendix Table B3 column (4) provides further details on the fertility reconstruction, which finds no evidence that extreme precipitation reduces fertility in urban areas. This result further supports our focus on rural fertility as the relevant margin of adjustment.

[Insert Table 10 about here]

7.3 Specification and Sample Checks

We next examine whether the baseline estimate is sensitive to alternative specification choices, local development trends, inference, or the inclusion of geographically unusual counties and recorded flood episodes. These checks address the concern that the estimated relationship might reflect differential local trends, correlated economic development, a particular clustering choice, or a small set of exceptional locations rather than extreme precipitation itself.

Figure 3 changes one feature of the baseline specification at a time. The first two exercises address potential confounding from local trends and development by adding prefecture-level city-specific trends and two-year-lagged nighttime lights and population density. The third clusters standard errors at the prefecture-city level rather than the county level. The remaining exercises alter the estimation sample by excluding municipalities, omitting climatically distinctive provinces, and dropping county-years after the first recorded flood. The estimated coefficients remain negative in all six exercises. In particular, the coefficient remains negative after excluding counties that experienced floods, indicating that the baseline result is not driven by observations associated with documented flood exposure. These checks show that the negative fertility response is stable across alternative trend controls, observable measures of local development, inference choices, geographic restrictions, and the exclusion of recorded flood episodes.

[Insert Figure 3 about here]

8 Conclusion

As extreme precipitation becomes more frequent, its consequences may extend beyond agricultural production and income to fertility and the spatial and sectoral allocation of labor in rural China. This paper examines these demographic effects using a county-year panel covering 2000–2014, reconstructed from the 2015 1% National Population Sample Survey and matched to precipitation data. County-level fertility is linked to extreme precipitation in the preceding year, with county fixed effects and province-by-year fixed effects.

Our findings show that an extremely dry or wet year reduces next-year fertility by approximately three women giving birth per 1,000 rural registered-address women of reproductive age. The relationship is nonlinear, with fertility declining at both the dry and wet extremes of the county-specific SPI distribution. This pattern remains stable across alternative fertility measures, exposure definitions, specifications, clustering choices, and sample restrictions, indicating that the fertility response is concentrated in severe precipitation anomalies.

Household-level evidence points to agricultural income loss and labor migration as two important channels underlying this decline. Extreme precipitation reduces agricultural income and increases migration among working-age adults, both of which can constrain household fertility decisions. Self-reported physical health, in contrast, shows little response. Local adaptive capacity also appears to matter: fertility declines are generally smaller where reservoirs, mechanization, nonfarm opportunities, or transport access are greater, although these short-run differences are estimated imprecisely and may become more pronounced over longer horizons. The contrast with the reconstructed 1990s panel further highlights the importance of the surrounding rural economic environment. Extreme precipitation is associated with higher fertility in the 1990s and lower fertility during 2000–2014. As nonfarm employment and migration opportunities expanded, household adjustment to agricultural shocks increasingly involved income disruption and labor mobility, while the labor and insurance value of children became less central.

More broadly, these findings show that climate adaptation has an important demographic dimension. The fertility consequences of extreme precipitation depend on how rural households stabilize income, reallocate labor, and maintain household co-residence after adverse weather. This implies that adaptation policies affecting irrigation, agricultural insurance, income support, transport connectivity, and migration-related public services may shape demographic responses alongside their effects on agricultural production and livelihoods. A promising direction for future research is to combine longer fertility histories with policy-specific variation in these forms of adaptation to examine how short-run fertility responses accumulate into longer-term changes in childbearing.

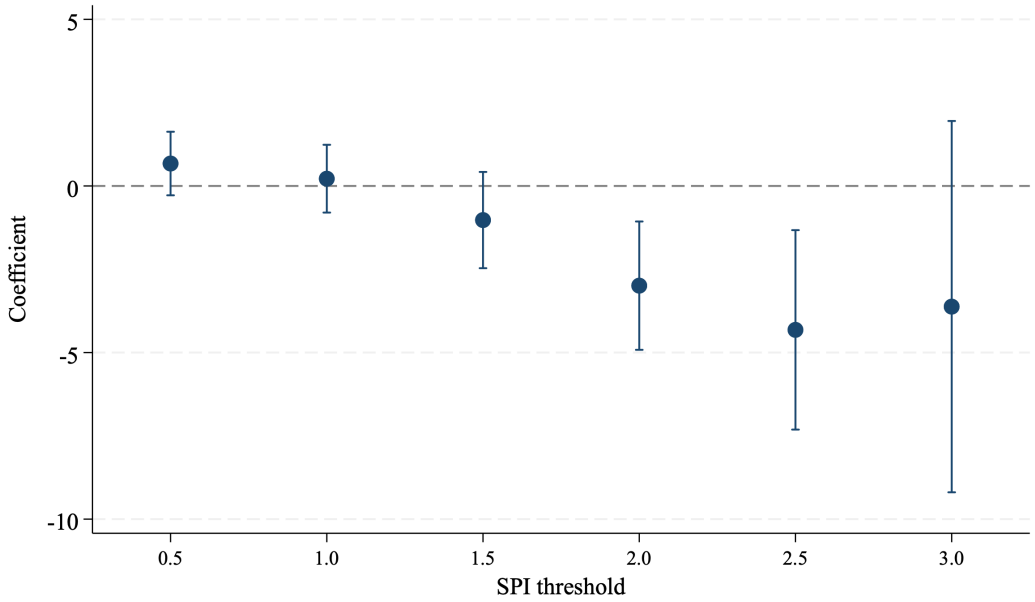


Fig. 2. Fertility Estimates at Alternative SPI Thresholds

Notes: Each point is the coefficient on one-year-lagged EP from a separate county-year regression using the 2000–2014 rural fertility panel. EP equals one when the absolute annual SPI reaches or exceeds the threshold shown on the horizontal axis, so the thresholds are cumulative and combine extremely dry and extremely wet conditions. Appendix B.1 Table B1 reports the corresponding means of the binary indicators in the baseline estimation sample. The dependent variable is the number of women giving birth per 1,000 rural registered-address women aged 20–49. Every specification includes county fixed effects, province-by-year fixed effects, and one-year-lagged standardized controls for temperature, specific humidity, and atmospheric pressure; standard errors are clustered at the county level. Vertical bars show 95% confidence intervals. Estimates at the strictest thresholds use fewer extreme county-years and are consequently less precise.

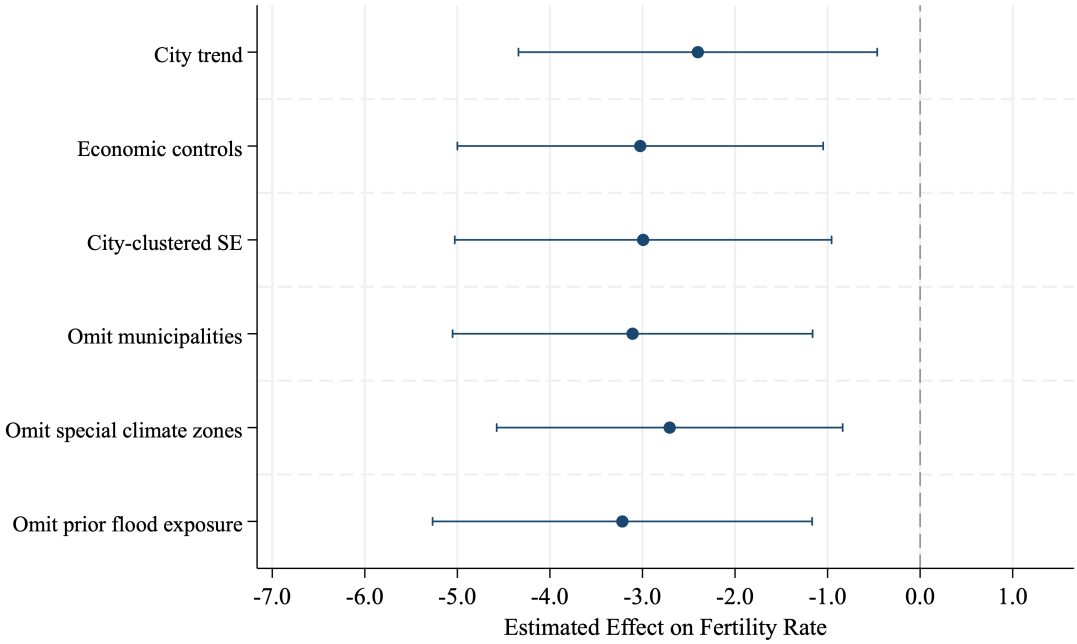


Fig. 3. Robustness: Specification and Sample Checks

Notes: Each point is the coefficient on lagged extreme precipitation from a separate county-year regression using the annual fertility panel, county-level precipitation, and climate controls. The six checks vary one feature at a time. They add prefecture-level city-specific trends, add two-year-lagged nighttime lights and population density, cluster standard errors at the prefecture-city level, omit municipalities, omit climatically distinctive provinces, and exclude county-years after the first recorded flood. Nighttime lights and population density are from (Wu et al., 2022) and LandScan, respectively. County and province-by-year fixed effects are included unless the exercise changes the fixed-effect structure, and every regression includes the baseline climate controls. Horizontal bars show 95% confidence intervals. The plotted objects are regression coefficients.

Table 1. Summary Statistics

	Obs.	Min.	Mean	Max.	SD
Panel A: Fertility Measures					
Fertility rate (rural)	34,722	0.00	36.12	1000.00	42.49
Fertility rate (urban)	32,893	0.00	29.08	1000.00	61.22
Panel B: Weather Variables					
Extreme precipitation indicator	34,722	0.00	0.05	1.00	–
Temperature (z-score)	34,722	-8.95	1.34	14.11	1.46
Specific humidity (z-score)	34,722	-3.71	0.77	4.38	1.00
Atmospheric pressure (z-score)	34,722	-3.85	-0.11	3.36	1.26
Panel C: County-Level Variables					
Nearby dam	34,711	0.00	0.57	1.00	–
Agricultural machinery	24,863	0.00	33.99	1512.00	38.15
County fiscal expenditure per capita (yuan)	23,770	76.12	428.03	4244.00	324.15
Primary-industry share	19,727	0.00	0.25	0.98	0.13
Secondary-industry share	19,648	0.00	0.42	0.99	0.17
Genealogies per 10,000 residents	15,540	0.01	2.81	108.09	9.44
Motorway density (m/km^2)	33,031	0.00	30.64	668.25	73.56
Panel D: Household and Individual Variables					
Head's gender	15,479	0.00	0.88	1.00	–
Head's age	15,479	18.00	46.74	59.00	8.29
Head's education (years)	15,469	0.00	7.96	19.00	3.28
Head's marital status	15,467	0.00	0.93	1.00	–
Child ratio	15,479	0.00	0.60	10.00	0.66
Total household income (thousand yuan)	15,058	0.00	45.42	622.20	58.53
Migrant	60,922	0.00	0.25	1.00	–
Self-reported health	58,077	0.00	0.67	1.00	–

Notes: Panel A reports county-year fertility measures as births per 1,000 women in the corresponding registered-address population. Panel B reports the one-year-lagged EP indicator and climate controls used in the preferred specification. EP equals one when the absolute annual SPI is at least two. Panel C reports county-level measures matched to the preferred baseline sample: nearby dams, machinery, county fiscal capacity, industrial structure, genealogy density, and motorway density. The dam indicator equals one when a GRanD dam is located in the county or a contiguous county. Genealogy density is measured as genealogies per 10,000 residents, and motorway density is measured in metres per square kilometre in 2000. Panel D reports CHFS household and individual variables extracted from the mechanism-analysis sample. Household-head and income rows retain households with a head aged 59 or younger; migration uses individuals aged 16–59; and health uses individuals aged 16 or older. Income is expressed in thousands of yuan. All county-year statistics use the preferred baseline regression sample; Obs. varies across rows because of missing values and because some external county measures have incomplete geographic coverage.

Table 2. Baseline: Effects Extreme Precipitation on Fertility

	Outcome: Fertility Rate				
	(1)	(2)	(3)	(4)	(5)
Extreme Dry	-3.162***		-3.132***		
	(1.155)		(1.155)		
Extreme Wet		-3.552***	-3.515***		
		(1.196)	(1.196)		
EP				-3.295***	-2.992***
				(0.828)	(0.982)
County FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	No
Province-by-year FE	No	No	No	No	Yes
Mean of Dep. Var.	36.117	36.117	36.117	36.117	36.117
Observations	34,722	34,722	34,722	34,722	34,722
Adj. R^2	0.204	0.204	0.204	0.204	0.216

Notes: The dependent variable is births per 1,000 rural registered-address women aged 20–49. Extreme Dry is defined as $SPI \leq -2$, Extreme Wet as $SPI \geq 2$, and EP equals one for either condition. Precipitation shocks and climate controls are lagged by one year. Standard errors clustered at the county level are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table 3. Mechanism: Effect of Extreme Precipitation on Household Income

	Outcome: Household Income			
	Any total income	Total income (IHS)	Any agricultural income	Agricultural income (IHS)
	(1)	(2)	(3)	(4)
EP	0.014	0.237	-0.092***	-0.897**
	(0.014)	(0.219)	(0.035)	(0.358)
Household controls	Yes	Yes	Yes	Yes
Climate controls	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Province-by-year FE	Yes	Yes	Yes	Yes
Observations	14,971	14,971	12,912	12,912
Adj. R^2	0.020	0.085	0.348	0.572

Notes: The data come from the 2011, 2013, and 2015 waves of the China Household Finance Survey, and the income outcomes refer to 2010, 2012, and 2014, respectively. The sample consists of households whose heads are younger than 60. Columns (1) and (3) indicate whether the household reports any total or agricultural income; columns (2) and (4) use the inverse hyperbolic sine (IHS) of the corresponding amount. Extreme precipitation is matched to the income year. Household controls include the household head's sex, age, education, and marital status and the share of household members younger than 16. All specifications include household and climate controls, county fixed effects, and province-by-year fixed effects. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4. Mechanism: Effect of Extreme Precipitation on Migration

	Outcome: Migration Indicator					
	All individuals			Unmarried individuals		
	All	Men	Women	All	Men	Women
	(1)	(2)	(3)	(4)	(5)	(6)
EP	0.032**	0.039**	0.026	0.059*	0.081**	0.031
	(0.016)	(0.017)	(0.018)	(0.034)	(0.041)	(0.046)
Household controls	Yes	Yes	Yes	Yes	Yes	Yes
Climate controls	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Province-by-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	60,702	31,464	29,238	15,742	9,496	6,242
Adj. R^2	0.577	0.538	0.623	0.458	0.434	0.495

Notes: The data come from the 2011, 2013, and 2015 waves of the China Household Finance Survey, and the migration outcomes refer to 2010, 2012, and 2014, respectively. The dependent variable equals one when the respondent was away from home for more than six months during the survey year. Columns (1)–(3) use all individuals aged 16–59 and report estimates for the pooled sample, men, and women. Columns (4)–(6) apply the same division to unmarried individuals aged 16–59. Extreme precipitation is matched to the migration year. All specifications include household and climate controls, county fixed effects, and province-by-year fixed effects. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5. Mechanism: Effect of Extreme Precipitation on Self-Reported Health

	Outcome: Self-Reported Health			
	Ages 16+	Ages 16–49	Women aged 16–49	Ages 60+
	(1)	(2)	(3)	(4)
EP	-0.033	-0.028	-0.043	-0.010
	(0.021)	(0.033)	(0.035)	(0.027)
Household controls	Yes	Yes	Yes	Yes
Climate controls	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Province-by-year FE	Yes	Yes	Yes	Yes
Observations	57,935	29,932	14,526	15,762
Adj. R^2	0.224	0.249	0.253	0.156

Notes: The data come from the 2011, 2013, and 2015 waves of the China Household Finance Survey, and the health outcomes refer to 2010, 2012, and 2014, respectively. The dependent variable equals one when the original five-category self-reported health measure takes a value from 3 to 5 and zero when it takes a value of 1 or 2. Columns (1)–(4) use adults aged 16 or above, adults aged 16–49, women aged 16–49, and adults aged 60 or above, respectively. Extreme precipitation is matched to the health year. All specifications include household and climate controls, county fixed effects, and province-by-year fixed effects. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6. Heterogeneous Effects of Extreme Precipitation by Adaptation Capacity

	Outcome: Fertility Rate			
	Reservoir	Mechanization	Fiscal capacity	Clan network
	(1)	(2)	(3)	(4)
EP	-4.367**	-4.363***	-1.513	-3.260**
	(1.971)	(1.612)	(1.129)	(1.424)
EP × Reservoir	2.148			
	(2.282)			
EP × Mechanization		2.679		
		(1.942)		
EP × Fiscal capacity			-2.177	
			(1.937)	
EP × Clan network				0.710
				(1.956)
County FE	Yes	Yes	Yes	Yes
Province-by-year FE	Yes	Yes	Yes	Yes
Mean of Dep. Var.	36.102	36.734	36.666	34.448
Observations	34,685	23,597	23,723	15,512
Adj. R^2	0.218	0.242	0.233	0.299

Notes: The dependent variable is births per 1,000 rural registered-address women aged 20–49. Each column estimates a pooled interaction model. The EP coefficient gives the estimated effect for the reference group, defined as counties without a nearby reservoir presence in 2000 in column (1), counties below the 2000 median of agricultural mechanization in column (2), counties below the 2000 median of fiscal expenditure per capita in column (3), and counties below the median of genealogy density in column (4). The interaction coefficient measures the difference in the EP effect between the corresponding groups. Nearby reservoir identifies counties with a GRanD dam in the county or a contiguous county in 2000. High mechanization is defined using the cross-county median in 2000. High fiscal capacity is defined using the cross-county median of county fiscal expenditure per capita in 2000. Dense clan network is defined by an above-median number of genealogies per 10,000 residents. The genealogy measure is compiled by 2014 from the Shanghai Library China Genealogy Knowledge Service Platform and normalized by the 1990 county population. Climate controls and province-by-year fixed effects are fully interacted with the relevant condition. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7. Heterogeneous Effects by Industrial Structure and Mobility Conditions

	Outcome: Fertility Rate		
	Primary share	Secondary share	Motorway
	(1)	(2)	(3)
EP	-1.899	-3.619**	-3.046***
	(1.349)	(1.475)	(1.112)
EP × Primary	-1.883		
	(1.951)		
EP × Secondary		1.987	
		(1.977)	
EP × Motorway			1.092
			(2.363)
County FE	Yes	Yes	Yes
Province-by-year FE	Yes	Yes	Yes
Mean of Dep. Var.	36.739	36.688	35.863
Observations	23,751	23,737	33,026
Adj. R^2	0.234	0.236	0.229

Notes: The dependent variable is births per 1,000 rural registered-address women aged 20–49. Each column estimates a pooled interaction model. In columns (1) and (2), the interaction term reports the differential EP effect in counties whose 2000 primary- or secondary-industry share is at or above the sample median. Column (3) defines motorway access as positive county motorway density in 2000. Climate controls and province-by-year fixed effects are fully interacted with the relevant condition. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8. Heterogeneous Effects of Extreme Precipitation by Period

	Outcome: Fertility Rate		
	1990s panel		2000–2014 panel
	$\|SPI\| \geq 1.5$	$\|SPI\| \geq 2.0$	$\|SPI\| \geq 2.0$
	(1)	(2)	(3)
EP	1.335*	3.201**	-2.992***
	(0.755)	(1.354)	(0.982)
Controls	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	No
Province-by-year FE	No	No	Yes
Observations	13,793	13,793	34,722
Adj. R^2	0.461	0.461	0.216

Notes: The dependent variable is the number of women giving birth per 1,000 women in the relevant fertility-reconstruction sample. Columns (1)–(2) use the retrospective 1990–1999 panel and define EP using $|SPI| \geq 1.5$ and $|SPI| \geq 2.0$, respectively, following Tian et al. (2026). Column (3) uses the main 2000–2014 panel and the baseline threshold of $|SPI| \geq 2.0$. All specifications include the baseline climate controls and county fixed effects. Columns (1)–(2) include year fixed effects, whereas column (3) includes province-by-year fixed effects. The table compares estimates across periods and does not report a pooled test of a structural break. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9. Robustness: Alternative Precipitation Measures

	Outcome: Fertility Rate			
	Z-score	SPI 1.5	SPI 2.5	Quadratic
	(1)	(2)	(3)	(4)
Extreme z-score	-2.268**			
	(0.980)			
Extreme SPI 1.5		-1.024		
		(0.737)		
Extreme SPI 2.5			-4.320***	
			(1.526)	
SPI				0.231
				(0.394)
SPI squared				-0.304*
				(0.169)
County FE	Yes	Yes	Yes	Yes
Province-by-year FE	Yes	Yes	Yes	Yes
Mean of Dep. Var.	36.117	36.117	36.117	36.117
Observations	34,722	34,722	34,722	34,722
Adj. R^2	0.216	0.216	0.216	0.216

Notes: The dependent variable is births per 1,000 rural registered-address women aged 20–49. Column (1) defines an extreme precipitation year using an annual rainfall z-score with an absolute value exceeding two. Columns (2)–(3) use alternative cumulative SPI thresholds of 1.5 and 2.5 in absolute value, respectively. Column (4) includes continuous SPI and its squared term to examine the nonlinear relationship between precipitation anomalies and fertility. Annual precipitation measures and climate controls are lagged by one year. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 10. Robustness: Alternative Outcomes and Exposure Windows

	Outcome: Fertility Rate			
	Ages 15–49	Birth ind.	Growing-season	Alt. panel
	(1)	(2)	(3)	(4)
Extreme precipitation	-3.079***	-0.002***	-1.804*	-1.849**
	(0.931)	(0.001)	(0.957)	(0.883)
County FE	Yes	Yes	Yes	Yes
Province-by-year FE	Yes	Yes	Yes	Yes
Mean of Dep. Var.	35.544	0.036	36.117	33.261
Observations	34,756	2,279,627	34,722	23,003
Adj. R^2	0.212	0.008	0.216	0.233

Notes: Column (1) defines the outcome as the annual number of women giving birth per 1,000 rural registered-address women aged 15–49. Column (2) uses a mother-year panel in which the dependent variable equals one if a woman gives birth in that year; multiplying the coefficient by 1,000 expresses the estimate as a change per 1,000 women. Column (3) uses the annual fertility rate for women aged 20–49 and constructs SPI from March–September growing-season precipitation. Column (4) combines fertility histories reconstructed from the 2010 Population Census for 2000–2009 with those reconstructed from the 2015 1% National Population Sample Survey for 2010–2014. Extreme precipitation equals one when $|\text{SPI}| \geq 2$. Annual precipitation measures are lagged by one year. All specifications include the corresponding climate controls, county fixed effects, and province-by-year fixed effects. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the authors used OpenAI's ChatGPT to assist with language editing, improving clarity and presentation, and checking consistency across the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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A Data and Variable Construction

This section documents the construction of the main outcomes, exposures, controls, and auxiliary variables. We describe measures reconstructed from household rosters, matched from external data, and defined in the CHFS questionnaires.

A.1 Fertility Histories and Age Definitions

We reconstruct the fertility sample from the 2015 1% National Population Sample Survey. We classify each respondent's registered address as rural or urban using administrative classification codes issued by the National Bureau of Statistics of China and retain women with rural registered addresses in the baseline sample. Because weather exposure is assigned using the registered county, we exclude women whose prolonged absence makes the registered address a poor proxy for preconception exposure. The baseline sample retains women who had never left or had been absent for no more than two years, following the duration categories reported in the survey.

The baseline outcome is defined using women aged 20–49 in each calendar year. Age eligibility is therefore reconstructed separately for every year from 2000 to 2014 rather than being fixed according to age at the time of the 2015 survey. Specifically, for each woman observed in the 2015 survey, we use her reported year of birth to calculate her age in year t and include her in the relevant denominator only if she was aged 20–49 in that year. The composition of the age-eligible population consequently changes across years as women enter and leave the 20–49 age range. The broader 15–49 range follows the convention used by the United Nations and national statistical agencies for fertility statistics and is examined as a robustness check.[24]

We reconstruct fertility histories for 2000–2014. Restricting the reconstruction to children aged 15 or younger at the time of the 2015 survey reduces omission of older children who may have left the household roster. We exclude births in 2015 because the November survey does not cover the full calendar year.

The survey does not provide a direct mother–child identifier. Following the general strategy in Tian et al. (2026), we infer mother–child links from each member's relationship to the household head, spousal relationships, and intergenerational parent–child relationships. Our linkage uses three adjacent generations comprising the household head's parents, the household head and spouse, and their children. We discard households in which a relationship path yields multiple plausible mothers. For each remaining link, the child's reported year and month of birth define a retrospective childbirth event for the potential mother. We then aggregate these events

[24]United Nations, Department of Economic and Social Affairs, Population Division, *World Population Prospects*, available at https://population.un.org/wpp/. World Health Organization, *Global Health Observatory*, available at https://www.who.int/data/gho.

by the mother's registered-address county and the child's year of birth.

For county c and year t, let W_{ct} denote the number of women observed in the 2015 survey who have rural registered addresses in county c and who, based on their reported year of birth, were aged 20–49 in year t. Let B_{ct} denote the number of women in this year-specific population with at least one inferred childbirth event in year t. The dependent variable is

$$FertilityRate_{ct} = \frac{B_{ct}}{W_{ct}/1{,}000}. \tag{2}$$

Thus, both the numerator and denominator are constructed using age eligibility in year t, rather than age at the time of the 2015 survey. The measure records the number of women giving birth per 1,000 women in the relevant year-specific population rather than the number of children born. A woman with multiple children in the same delivery is counted once. We apply the same procedure to the alternative age range, the urban registered-address sample, and the retrospective panel that combines the 2010 Population Census with the 2015 survey.

A.2 Precipitation and Climate Controls

The raw weather records come from NOAA's Global Surface Summary of the Day database. The records are spatially interpolated to a $0.1^\circ \times 0.1^\circ$ grid using the five nearby stations, converted to raster format, and averaged within the county to form county-level daily, monthly, and annual panels from 1973 onward. We use observations from 1973–2014 in the annual analysis. When multiple source records are available for a county-day, we first average them by county and date and then aggregate precipitation to annual totals.

We construct a county-specific Standardized Precipitation Index (SPI) using the Gamma-distribution approach of McKee et al. (1993) and 1973–1997 as the historical reference period. The fitted distribution and reference-period length affect SPI estimates (Guttman, 1999). Our reference period contains 25 annual observations for each county. Let x_{ct} denote annual precipitation in county c and year t. For each county, positive historical precipitation observations are fitted to a Gamma distribution,

$$g_c(x) = \frac{x^{\alpha_c - 1} e^{-x/\beta_c}}{\beta_c^{\alpha_c} \Gamma(\alpha_c)}, \qquad x > 0, \tag{3}$$

where α_c and β_c are county-specific shape and scale parameters. If $G_c(x)$ denotes the associated cumulative distribution and q_c is the historical frequency of zero precipitation, the adjusted cumulative probability is

$$H_c(x) = q_c + (1 - q_c) G_c(x). \tag{4}$$

The SPI is then

$$SPI_{ct} = \Phi^{-1}\left[H_c(x_{ct})\right], \tag{5}$$

where $\Phi^{-1}(\cdot)$ is the inverse standard normal distribution. Negative values indicate precipitation below the fitted historical median and positive values indicate precipitation above it. The transformation accommodates skewed rainfall distributions and places county-specific tail probabilities on a comparable scale (Guttman, 1999; Wang et al., 2022).

The baseline exposure is $EP_{ct} = 1$ when $|SPI_{ct}| \geq 2$, which combines the extremely dry and extremely wet tails. We also estimate separate dry and wet indicators, alternative SPI thresholds, conventional precipitation z-scores, and percentile-based precipitation indicators. The wet-side threshold captures an annual precipitation anomaly and may reflect unusually abundant rainfall as well as damaging rainfall. Annual regressions use a one-year lag.

For the non-precipitation controls, we average daily temperature, specific humidity, and atmospheric pressure to the county-year level and standardize each variable relative to its county-specific 1973–1997 mean and standard deviation. The standardized controls capture local weather anomalies rather than persistent cross-county climatic differences. Temperature measures thermal conditions relevant to agricultural production (Chen et al., 2016; Chen and Gong, 2021). Specific humidity captures atmospheric moisture, which can affect estimated climate effects on Chinese crop yields (Zhang et al., 2017). Atmospheric pressure provides a parsimonious control for co-occurring synoptic conditions, consistent with related Chinese county-level work (Li et al., 2020).

Figure A1 shows considerable cross-county variation in mean SPI over 2000–2014: negative averages are widespread across northern and southwestern China, while positive averages appear in parts of the Tibetan Plateau and the southeastern coast. Figures A2a and A2b show that changes in daily precipitation conditions are also spatially uneven. Dry days increased most visibly in the southwest but declined across many northern and central counties, whereas increases in days above the historical wet-day 95th percentile are concentrated in many southern and southwestern counties, with localized declines in central and northeastern areas.

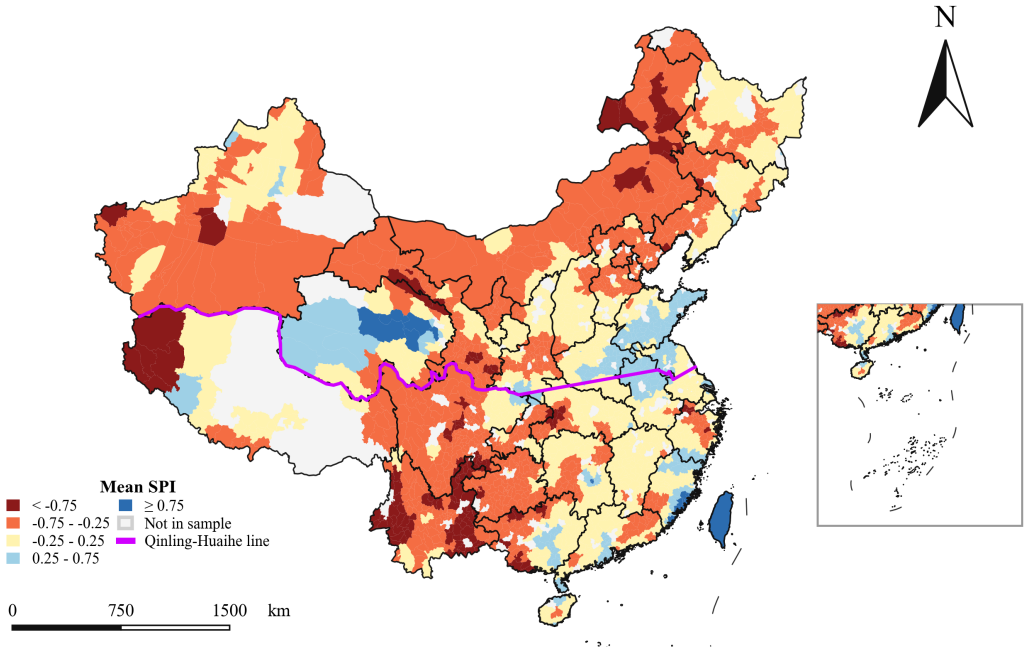
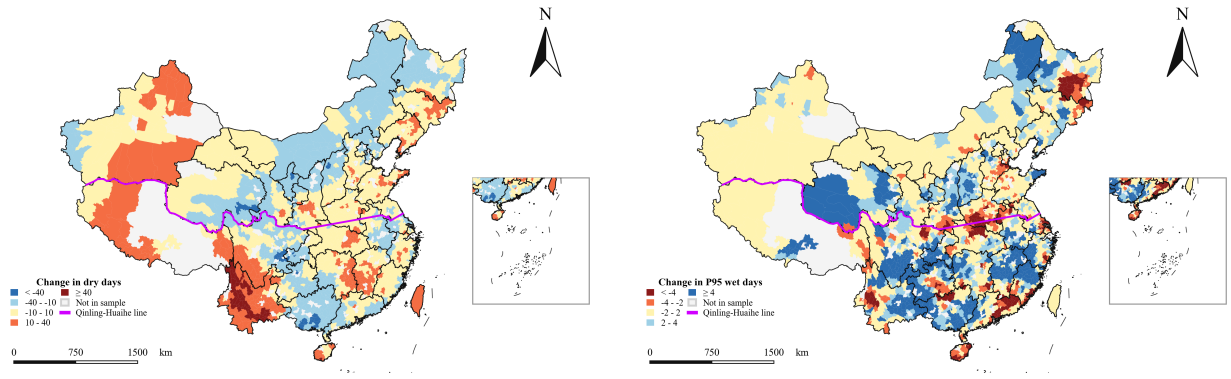


Fig. A1. Average SPI Across Counties, 2000–2014

Notes. The map uses county-level annual SPI values constructed from NOAA GSOD precipitation observations and plots the county-specific mean over 2000–2014. SPI is calculated relative to each county's 1973–1997 historical precipitation distribution. The figure provides a descriptive summary of spatial variation in SPI.



(a) Change in dry days

(b) Change in wet days

Fig. A2. Changes in Dry and Wet Days, 2000–2014

Notes. The maps use daily county-level precipitation observations for 1973–2014. The left panel reports the change between 2000 and 2014 in the annual number of dry days, defined as days with precipitation below 1 mm. The right panel reports the corresponding change in days with precipitation above each county's historical wet-day 95th percentile. County-specific thresholds are calculated from wet days during 1973–1997, where a wet day has precipitation of at least 1 mm. The maps summarize changes in daily precipitation conditions. County-level variables are mapped using five classes. Counties shown in gray are not included in the sample. County- and provincial-level administrative boundaries are based on Administrative Boundary Maps of China in 2014, accessed through the University of California, Berkeley Library. The ten-dash line in the South China Sea is based on administrative-boundary data provided by the National Platform for Common Geospatial Information Services (Tianditu). The source map carries review number GS (2024) 0650.

A.3 County-Level Variables

County-level agricultural output, rural income, agricultural machinery, labor, and industrial indicators come from the county statistical yearbook series. The resulting county-year panel supplies the crop-output, machinery, and industry measures used in the supplementary analysis. Monetary and production quantities enter in IHS form where indicated. Shares, including agricultural GDP as a share of primary-sector GDP and the secondary-industry share, remain in levels.

The specification checks also use nighttime-light intensity from the harmonized Chinese DMSP-OLS-like series of Wu et al. (2022) and population density from LandScan.[25] We collapse repeated county-year records to one observation, retain the county-year mean of each measure, and transform both variables using the inverse hyperbolic sine. We match them to the fertility panel by county code and year and include them with a two-year lag in the corresponding robustness specification.

Dam locations are drawn from the Global Reservoir and Dam Database (GRanD) (Lehner et al., 2011). Related evidence considers flood-control dams as a potential moderator of the educational effects of flooding (Huang and Dong, 2025). We construct the reservoir presence indicator from irrigation dams recorded as operating by 2000. The indicator equals one when at least one GRanD dam is located in the county or a contiguous county. This spatial extension captures water-control infrastructure that may affect agricultural conditions beyond the administrative boundary containing the dam. The mechanization split is based on whether a county's 2000 agricultural-machinery measure is above the cross-county median. For fiscal capacity, we divide 2000 county fiscal expenditure by year-end population and classify counties above the cross-county median as high capacity. This measure captures baseline differences in local public capacity.

Two additional cross-sectional measures capture clan networks and mobility access. We draw genealogy records from the Shanghai Library's China Genealogy Knowledge Service Platform, which consolidates catalogue records held by Chinese and overseas institutions.[26] We retain records compiled by 2014 with identifiable county locations and compilation dates. We count genealogies by county and divide by the 1990 county resident population before multiplying by 10,000. The resulting measure records genealogies per 10,000 residents. The matched inventory covers 1,201 counties with identifiable county information, and the descriptive and interaction analyses use this corresponding sample.

Motorway measures come from the county panel of the China Surface Transport System

[25] LandScan data and documentation are available from the Oak Ridge National Laboratory at https://landscan.ornl.gov/.

[26] The platform and its catalogue are available at China Genealogy Knowledge Service Platform.

Table A1. Data Sources for Auxiliary Variables

Variable	Source
Precipitation, temperature, specific humidity, and pressure	NOAA GSOD; county weather panel
Household income, migration, and self-reported health	China Household Finance Survey (CHFS), SWUFE; Gan et al. (2016)
Agricultural GDP and output, rural income, labor, machinery, fiscal expenditure and population, and industry shares	County statistical yearbooks
Dam locations	Global Reservoir and Dam Database (GRanD)
Clan-network density	Shanghai Library China Genealogy Knowledge Service Platform; county-level aggregation and normalization follow the procedure described in Appendix A.3
Motorway length and density (2000)	Davis et al. (2025); China Surface Transport System Database
Nighttime lights and population density	Wu et al. (2022); LandScan

Note: This table lists the data sources used for the mechanism, heterogeneity, and specification analyses. The linked source names provide access to the relevant data product or documentation. Detailed variable definitions and processing steps are reported in Appendix A.2 and A.3. The CHFS row covers the household-level income, individual migration, and self-reported health outcomes used in the mechanism analysis. GRanD dams located in contiguous counties are included in the infrastructure analysis. The genealogy measure counts county-coded records compiled by 2014 and normalizes them by 1990 county resident population; records without county identifiers are excluded, so its interaction analysis uses the matched coverage sample. Motorway access is defined from positive 2000 motorway density, while the table reports the continuous density measure. Fiscal capacity is measured as 2000 county fiscal expenditure per capita, calculated from county fiscal expenditure and year-end county population.

Database compiled by Davis et al. (2025).[27] We use the 2000 observation and measure total motorway length within a county and motorway length per square kilometre. The interaction analysis defines pre-existing motorway access as positive motorway density, while the descriptive table reports continuous density in metres per square kilometre. We match all cross-sectional measures to the annual fertility panel by county code. Table A1 summarizes the external data sources used in the mechanism, heterogeneity, and specification analyses.

[27]The data documentation is available at the China Surface Transport System Database readme.

B Additional Empirical Results

This section reports supplementary evidence on alternative definitions of extreme precipitation, heterogeneity by employment and residence status, and agricultural, industrial, income, and migration adjustment. These results complement the main estimates and clarify the scope and interpretation of the baseline relationship.

B.1 Alternative Definitions of Extreme Precipitation

Table B1 reports the exposure share under alternative SPI thresholds. A threshold of 0.5 classifies 61.2% of baseline county-years as extreme, while a threshold of 1.0 classifies 31.2%. These definitions capture relatively broad deviations from local precipitation conditions. The preferred threshold of 2.0 selects 5.2% of county-years and corresponds to the conventional extreme SPI category while retaining enough tail observations for county-year estimation. Thresholds of 2.5 and 3.0 reduce the exposure share to 1.6% and 0.5%, respectively, which is reflected in wider confidence intervals at the strictest thresholds in Figure 2. The 2.0 threshold therefore balances the identification of unusually severe precipitation with sufficient empirical support.

As an additional nonparametric check, Appendix Table B2 defines extreme precipitation directly from the county-specific historical distribution of annual cumulative rainfall. The cutoffs range from outside the 20th–80th percentile interval to outside the 3rd–97th percentile interval. The latter is designed to approximate the tails identified by $SPI \geq 2$.[28] Accordingly, we use the county-specific 3rd and 97th percentiles of annual precipitation over 1973–1997 as a nonparametric approximation to the same tails (McKee et al., 1993). The two measures are not identical, however, because SPI relies on a fitted distribution, whereas the empirical-percentile measure is calculated directly from a limited number of historical annual observations.

The percentile-based coefficients are close to zero under the broader definitions and become more negative as the tails are narrowed to the 5th–95th and 3rd–97th percentiles, although they remain imprecisely estimated. This lower precision need not contradict the SPI results. With only about 25 annual observations in the historical reference period, the most extreme empirical percentiles are close to the county-specific sample minimum and maximum and are therefore estimated with considerable sampling uncertainty. Such uncertainty may alter which county-years are classified as extreme, while the binary percentile indicators also discard information about the severity of deviations beyond the cutoff. Both features can attenuate the estimates and increase their standard errors. SPI, by contrast, uses the fitted historical distribution to place

[28] Because SPI maps the fitted county-specific precipitation distribution into a standard normal distribution, $SPI = 2$ corresponds to a cumulative probability of approximately 0.977, while $SPI = -2$ corresponds to approximately 0.023. Thus, $SPI \geq 2$ identifies observations in roughly the upper or lower 2.3 percent of the fitted historical distribution.

Table B1. Means of Extreme Precipitation Indicators at Alternative SPI Thresholds

Absolute SPI threshold	Mean (share classified as extreme)
0.5	0.612
1	0.312
1.5	0.136
2	0.052
2.5	0.016
3	0.005

Notes: Each row defines the one-year-lagged extreme precipitation indicator as equal to one when the absolute annual SPI reaches or exceeds the threshold shown in the first column. The table uses the 34,722 county-year observations retained in the preferred baseline regression sample. Because the indicator is binary, its mean is also the share of county-years classified as extreme under the corresponding threshold. The regressions use county and province-by-year fixed effects, lagged climate controls, and standard errors clustered at the county level.

precipitation anomalies on a common probability scale across counties.

The percentile-based results provide weaker but broadly compatible evidence. These findings do not establish 2 as a uniquely optimal threshold, but they support retaining $SPI \geq 2$ as the preferred definition: it has a conventional meteorological interpretation, facilitates comparisons across counties with different precipitation distributions, and identifies severe events without reducing the treatment group to an excessively small number of observations.

Next, we try the continuous-SPI specification underlying column (4) of Table 9:

$$Fertility_{ct} = \beta_1 SPI_{c,t-1} + \beta_2 SPI^2_{c,t-1} + X_{c,t-1}\gamma + \alpha_c + \lambda_{pt} + \varepsilon_{ct}. \tag{6}$$

Here, $SPI_{c,t-1}$ is the one-year-lagged county-specific precipitation index, $X_{c,t-1}$ contains the lagged standardized climate controls, and the fixed effects and clustering follow the baseline specification. The estimates are $\widehat{\beta}_1 = 0.231$ and $\widehat{\beta}_2 = -0.304$. The negative quadratic coefficient implies an inverted U-shaped relationship. The fitted contribution of precipitation is highest near the turning point, $-\widehat{\beta}_1/(2\widehat{\beta}_2) \approx 0.38$, and declines as SPI moves toward either tail. Figure B1 plots this two-sided pattern with pointwise 95% confidence intervals.

B.2 Employment and Residence Heterogeneity

Table B3 examines whether the fertility response varies with household exposure to agriculture. The coefficient is negative in all four columns, with an estimate of -2.310 for agricultural households. Estimates for mixed-employment, nonagricultural, and urban-hukou households are

Table B2. Robustness: Alternative Historical Percentile Definitions of Extreme Precipitation

	Outcome: Fertility Rate			
	p_{20}–p_{80}	p_{10}–p_{90}	p_{5}–p_{95}	p_{3}–p_{97}
	(1)	(2)	(3)	(4)
EP	-0.227	0.010	-0.766	-1.476
	(0.472)	(0.576)	(0.682)	(0.935)
Controls	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Province-by-year FE	Yes	Yes	Yes	Yes
Observations	34,722	34,722	34,722	34,722
Adj. R^2	0.215	0.215	0.215	0.216

Notes: The dependent variable is the number of women giving birth per 1,000 rural registered-address women aged 20–49. Each column defines EP using the indicated lower and upper percentiles of the county-specific 1973–1997 distribution of annual cumulative precipitation. For example, the p_{20}–p_{80} indicator equals one below the 20th percentile or above the 80th percentile. The indicators are evaluated over 1998–2014 and lagged by one year. All specifications include the baseline controls, county fixed effects, and province-by-year fixed effects. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

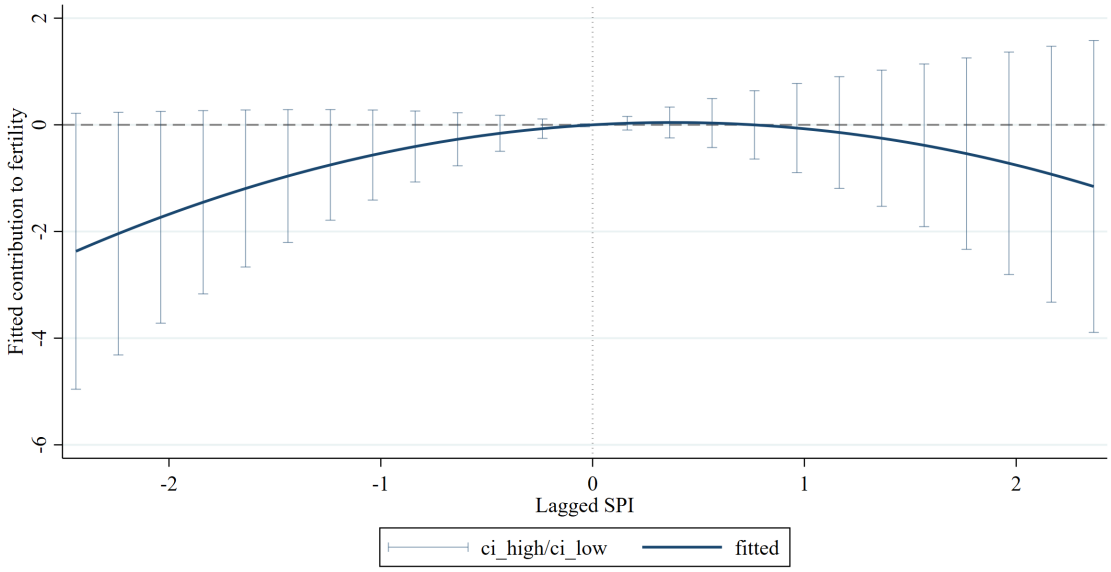


Fig. B1. Fitted Nonlinear Relationship Between SPI and Fertility

Notes. The figure uses the 34,722 county-year observations in the preferred baseline sample. It plots the fitted contribution $\widehat{\beta}_1 SPI + \widehat{\beta}_2 SPI^2$ from the continuous-SPI specification in Table 9, evaluated from the first to the 99th percentile of lagged SPI. Vertical bars show pointwise 95% confidence intervals calculated with the delta method using the estimated variance-covariance matrix. The plotted curve reports the contribution of the linear and quadratic SPI terms.

Table B3. Heterogeneous Effects of Extreme Precipitation by Household Employment Structure and Hukou Status

	Outcome: Fertility Rate			
	Household employment structure			Hukou status
	Agriculture	Mixed	Non-agriculture	Urban
	(1)	(2)	(3)	(4)
EP	-2.310*	-2.390	-1.563	-1.436
	(1.391)	(3.295)	(2.371)	(1.511)
Controls	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Province-by-year FE	Yes	Yes	Yes	Yes
Observations	33,050	28,606	33,045	32,921
Adj. R^2	0.145	0.053	0.063	0.032

Notes: The dependent variable is the number of women giving birth per 1,000 registered-address women aged 20–49 in the indicated group. Columns (1)–(3) use agricultural, mixed-employment, and nonagricultural households, respectively; household employment type is classified from the occupations of the household head and spouse. Column (4) uses women with urban hukou. All specifications include the baseline controls, county fixed effects, and province-by-year fixed effects. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

less precise. Only the agricultural-household estimate is marked as statistically significant; estimates in the mixed-employment, nonagricultural, and urban-hukou samples are less precise.

B.3 Agricultural, Industrial, and Income Adjustment

Table B4 supplements the household-income evidence with county-level outcomes. Panel A shows no detectable contemporaneous change in aggregate agricultural GDP or its sectoral shares. Panel B shows no detectable change in grain or oil-crop output, while cotton output rises. Panel C shows an increase in industry's share of secondary-sector output, while neither county-level industrial output nor the log of average household agricultural income changes significantly. Together, these results suggest that, at the county level, extreme precipitation induces adjustments in households' agricultural production and employment choices.

B.4 Migration Supplement

Table B5 restricts the migration analysis to married working-age adults, for whom sustained absence can affect co-residence and opportunities for conception. The pooled coefficient of 0.027 implies a 2.7-percentage-point increase in the probability of being away from home for more than

Table B4. Mechanism: County-Level Agricultural and Industrial Adjustment

Panel A: Agricultural Production	Agricultural GDP	Agricultural share of primary GDP	Agricultural share of total GDP
EP	0.019	0.006	−0.000
	(0.015)	(0.007)	(0.002)
Obs.	3,984	3,984	3,984
Adj. R^2	0.996	0.977	0.983
Panel B: Agricultural Adaptation	Grain output	Cotton output	Oil-crop output
EP	−0.004	0.080**	0.022
	(0.007)	(0.040)	(0.015)
Mean of Dep. Var.	12.466	6.728	8.997
Obs.	27,968	13,684	28,702
Adj. R^2	0.964	0.942	0.917
Panel C: Industrial Adjustment	Industrial GDP	Secondary-industry share	Rural income
EP	−0.006	0.247**	−0.001
	(0.016)	(0.112)	(0.015)
Obs.	28,947	23,590	6,527
Adj. R^2	0.942	0.236	0.945
Controls	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Province-by-year FE	Yes	Yes	Yes

Note: This table uses county-year agricultural and economic data. Panel A reports the IHS of agricultural GDP and agricultural-GDP shares in primary and total GDP. Panel B reports the IHS of grain, cotton, and oil-crop output. Panel C reports the IHS of industrial GDP and rural income and the secondary-industry share. EP equals one when annual SPI is at or below −2 or at or above 2. Panel A uses contemporaneous EP; Panels B and C use EP lagged by one year. Monetary and production outcomes enter in IHS form, whereas shares remain in levels. All specifications include standardized temperature, specific humidity, and atmospheric pressure, county fixed effects, and province-by-year fixed effects. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B5. Mechanism: Working-Age Migration among Married Individuals

	Outcome: Migration Indicator		
	All	Men	Women
	(1)	(2)	(3)
EP	0.027*	0.028	0.026
	(0.016)	(0.017)	(0.018)
Controls	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Province-by-year FE	Yes	Yes	Yes
Observations	44,869	21,923	22,945
Adj. R^2	0.648	0.612	0.684

Notes: The data come from the 2011, 2013, and 2015 waves of the China Household Finance Survey, and the migration outcomes refer to 2010, 2012, and 2014, respectively. The dependent variable equals one when the respondent was away from home for more than six months during the survey year. The sample consists of married individuals aged 16–59. Column (1) pools men and women; columns (2) and (3) use men and women separately. Extreme precipitation is matched to the migration year. All specifications include household and climate controls, county fixed effects, and province-by-year fixed effects. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

six months. The sex-specific coefficients are similar in magnitude but less precise. The pooled estimate is consistent with migration-related spousal separation as one margin of household adjustment.

B.5 Geographic and Agroclimatic Checks

Table B6 examines whether the fertility response varies across broad geographic and terrain conditions. In column (1), the EP coefficient is -3.427 for southern counties, while the positive interaction with the North indicator implies a somewhat smaller decline in northern counties. The interaction coefficient is not statistically significant, providing no evidence of a differential response between northern and southern counties. Column (2) shows a similar pattern across terrain conditions, with a negative EP coefficient in flatter counties and an interaction of only 0.103 for steeper terrain. Overall, the estimates remain negative across these geographic groups, while the interaction terms indicate no statistically significant heterogeneity by region or terrain slope.

Table B6. Heterogeneous Effects of Extreme Precipitation by Geographic Conditions

	Outcome: Fertility Rate	
	North	Terrain slope
	(1)	(2)
EP	-3.427***	-3.029*
	(1.230)	(1.545)
EP × North	1.192	
	(2.096)	
EP × Steeper terrain		0.103
		(2.057)
County FE	Yes	Yes
Province-by-year FE	Yes	Yes
Mean of Dep. Var.	36.117	36.122
Observations	34,722	34,717
Adj. R^2	0.215	0.213

Notes: The dependent variable is births per 1,000 rural registered-address women aged 20–49. Each column estimates a pooled interaction model. The EP coefficient applies to the reference group, while the interaction coefficient measures the differential EP effect for northern counties or counties with above-median terrain slope. Counties are classified as northern or southern according to the Qinling–Huaihe Line. Terrain slope is derived from the NASADEM Global Digital Elevation Model accessed through OpenTopography. NASADEM reprocesses elevation data collected during the 2000 Shuttle Radar Topography Mission using updated algorithms and auxiliary elevation sources and provides global elevation data at approximately 30-meter spatial resolution. We calculate terrain slope from the elevation surface and average slope within each county. Counties with mean slope above the cross-county median are classified as having steeper terrain. Climate controls and province-by-year fixed effects are fully interacted with the corresponding heterogeneity indicator. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.